

1 INTRODUCTION

1.1 Overview of the Project

The project, titled “Deep Learning-Based Brain Tumor Detection and Segmentation with an Integrated Doctor–Patient Portal,” focuses on addressing the challenges associated with accurate and efficient diagnosis of brain tumors. Brain tumors are among the most critical neurological disorders affecting the central nervous system and are caused by the abnormal and uncontrolled growth of cells within the brain. These tumors may be classified as benign or malignant, with malignant tumors posing significant risks to human life. Early detection and accurate diagnosis are considered essential for effective treatment planning and improving patient survival rates. Magnetic Resonance Imaging (MRI) is widely used in medical diagnostics due to its ability to produce high-resolution images with excellent soft tissue contrast, enabling detailed visualization of brain structures.

However, the interpretation of MRI scans is considered a complex and time-intensive process that depends on the expertise of radiologists. Variations in tumor characteristics and subtle abnormalities may lead to inconsistencies in diagnosis. To address these challenges, deep learning-based approaches have been introduced to support automated medical image analysis, where advanced segmentation models are capable of learning complex spatial patterns from MRI data, thereby improving detection accuracy and reducing dependency on manual analysis.

In the project, multiple deep learning models such as CNN, U-Net, and Attention U-Net were implemented and evaluated to identify the most suitable model for tumor segmentation. Based on comparative analysis using performance metrics such as Dice Score and Accuracy, *Attention U-Net is selected as the final model* due to its improved segmentation performance and ability to focus on relevant tumor regions.

In addition to automated tumor detection, a complete web-based Doctor–Patient Portal with an administrative module is developed to provide an integrated healthcare solution. The portal enables patients to upload MRI scans, view diagnostic results, and access medical reports, while doctors are provided with facilities to review segmentation outputs and offer medical recommendations. The administrative module is responsible for managing system operations and data.

The system also incorporates an automated report generation feature, where detailed tumor detection results, including tumor type, segmented output, tumor location, and evaluation metrics such as Dice Score and Accuracy, are generated in the form of downloadable PDF reports for clinical reference. This integration of artificial intelligence with a web-based platform ensures efficient data management, improved accessibility, and enhanced communication among users, thereby delivering a comprehensive and intelligent healthcare system.

1.2 Background of the Study

Magnetic Resonance Imaging (MRI) is widely used for brain tumor diagnosis due to its ability to provide high-resolution images with superior soft tissue contrast. Different MRI modalities such as T1, T1-contrast enhanced (T1c), T2, and Fluid Attenuated Inversion Recovery (FLAIR) provide complementary information about tissue characteristics, with FLAIR being particularly effective in highlighting abnormal regions and edema. However, the interpretation of MRI scans is considered a complex and time-intensive task that depends on the expertise of radiologists, which may lead to variability and inconsistencies in diagnosis.

With the advancement of Artificial Intelligence (AI), deep learning techniques have been increasingly applied in medical image analysis to improve diagnostic efficiency. Among these, U-Net architecture is widely used for biomedical image segmentation due to its encoder–decoder structure, while Attention U-Net further enhances segmentation performance by focusing on

relevant regions and suppressing background noise. The availability of standardized datasets such as the BraTS dataset has enabled the development of reliable models capable of learning complex spatial and intensity variations in MRI data.

In addition to AI-based advancements, modern healthcare systems have increasingly adopted web-based technologies for managing patient data and clinical workflows. However, many existing systems lack proper integration between diagnostic tools and healthcare management, resulting in inefficiencies and communication gaps. To address these limitations, the proposed system integrates deep learning-based tumor detection with a web-based Doctor–Patient portal, enabling centralized data management, improved accessibility, and efficient interaction among patients, doctors, and administrators.

1.3 Existing System

In the existing healthcare environment, brain tumor detection is primarily performed through the manual analysis of Magnetic Resonance Imaging (MRI) scans by radiologists. Although MRI provides high-resolution images with excellent soft tissue contrast, the interpretation process is time-consuming and highly dependent on the expertise of medical professionals, which may lead to variability and inconsistencies in diagnosis. Differences in tumor size and location make accurate detection difficult, particularly in the case of small or irregular tumor regions such as enhancing tumors. These challenges may result in misdiagnosis or delayed detection, adversely affecting treatment planning and patient survival rates.

Conventional systems do not provide precise tumor localization or pixel-level segmentation, such as bounding box identification or detailed delineation of tumor regions, which are essential for accurate clinical assessment. Furthermore, existing approaches rely mainly on visual inspection and do not utilize advanced deep learning techniques such as Convolutional Neural Networks (CNN), U-Net, and Attention U-Net for automated tumor detection and segmentation. As a result, these

systems lack the ability to perform accurate and consistent segmentation of tumor regions. Standard evaluation metrics such as Dice Score, Accuracy, Precision, and Recall are also not widely applied, limiting the reliability and consistency of diagnostic results.

Another major limitation is the lack of automation and integration within healthcare systems. Patient records, diagnostic reports, and appointment scheduling are often managed through separate platforms, resulting in inefficiencies, data redundancy, and communication gaps between patients, doctors, and administrators. Additionally, these systems do not support automated report generation or centralized data access, which further reduces system efficiency and usability. Although Artificial Intelligence (AI) has demonstrated significant potential in medical image analysis, its integration with real-time healthcare management systems remains limited.

Therefore, there is a need for an intelligent and integrated solution that combines deep learning-based tumor detection with a centralized Doctor–Patient portal to improve diagnostic accuracy, enhance system efficiency, and provide seamless healthcare services.

1.4 Proposed System

An intelligent and automated framework for brain tumor detection and segmentation is developed using deep learning methodologies, with primary emphasis on the Attention U-Net architecture. The model is designed to perform precise pixel-level segmentation by focusing on relevant tumor regions while suppressing background noise. Initially, multiple models such as CNN, U-Net, and Attention U-Net were implemented and evaluated to analyze their performance in tumor segmentation. Based on this comparative analysis, Attention U-Net demonstrated superior performance and achieved higher Dice Score values, and was therefore selected as the final model for the system. This attention mechanism

enhances the localization of tumor sub-regions such as Edema, Tumor Core, and Enhancing Tumor, thereby improving segmentation accuracy and reliability.

MRI data is preprocessed through operations such as normalization, resizing, noise reduction, and slice extraction to ensure standardized input for the model. The system is trained and validated using the BraTS dataset, which provides annotated multimodal MRI images. This enables the model to effectively learn variations in tumor size, shape, and intensity, while also supporting performance evaluation and comparison using metrics such as Dice Score.

In addition to the detection component, a web-based Doctor–Patient portal is implemented to facilitate system interaction and data management. MRI images are uploaded through the portal and processed via a Flask-based interface, and the generated segmentation outputs are stored in the system and made accessible through the portal and downloadable PDF reports. Patients are provided access to results and reports, doctors are enabled to review and analyze outputs, and administrators manage system operations and overall data flow. The system is developed using Python, TensorFlow, and Keras for deep learning, along with HTML, CSS, JavaScript, PHP, and MySQL for web development.

System performance is evaluated using metrics such as Dice Score, Accuracy, Precision, and Recall to assess segmentation quality. The Attention U-Net model achieved Dice Score values in the range of approximately 0.80-85, indicating improved overlap between predicted and ground truth tumor regions compared to other models. Data security is ensured through authentication and role-based access control. The system architecture is designed to be scalable, allowing future enhancements and improved deployment capabilities.

1.5 System Modules

The proposed system is structured into multiple modules to ensure efficient operation and management. The Patient Module enables users to upload MRI images, view results, download reports, and manage appointments. The Doctor Module allows medical professionals to access patient data, analyze MRI scans and segmentation results, and provide recommendations. The Admin Module manages system operations, including user control, data monitoring, and report handling. The Detection Module performs automated tumor segmentation using deep learning model such as Attention U-Net. The Database Module ensures secure storage and management of all system data using MySQL, and all modules are integrated within a centralized web-based Doctor–Patient portal for efficient healthcare delivery.

1.6 System Configuration

The system configuration defines the hardware and software environment required for the development and implementation of the proposed brain tumor detection and segmentation system. The system integrates deep learning techniques with a web-based Doctor–Patient portal to provide an efficient and intelligent healthcare solution. The web application is developed using HTML, CSS, and JavaScript for the frontend, and PHP for backend processing, and is deployed using WAMP Server. The portal enables functionalities such as MRI upload, result viewing, report access, and appointment management, ensuring smooth interaction among patients, doctors, and administrators.

The artificial intelligence module is implemented using Python with TensorFlow and Keras for automated tumor detection and segmentation. MRI images from the BraTS dataset are used for training and evaluation, with preprocessing techniques such as normalization and resizing performed using OpenCV. The system uses a MySQL database for secure data storage and management. Integration between the AI module and the web application is

achieved using the Flask framework, enabling real-time processing and communication. Development and testing are carried out using PyCharm, ensuring reliable system performance.

1.6.1 Hardware Requirements

The hardware requirements for the development and execution of the proposed system are specified to ensure efficient performance and smooth operation of both the web application and the artificial intelligence module. A system with adequate processing power and memory is required to handle deep learning computations and web-based functionalities.

In addition to the basic hardware configuration, the system requires a reliable storage and processing environment to efficiently manage large MRI datasets and trained model files. A high-performance CPU or GPU is recommended to accelerate deep learning model training and inference, particularly for handling complex architecture such as Attention U-Net. Sufficient RAM is essential to support simultaneous execution of the web server, database operations, and AI processing without performance degradation.

- *Operating Environment:* Windows-based system used for development and execution with WAMP Server installed.
- *Processor:* Intel Core i3 or above
- *RAM:* Minimum 8 GB
- *Storage:* Minimum 256 GB available space
- *Hard Disk:* 512 GB or above

1.6.2 Software Requirements

The software requirements include the necessary tools, frameworks, and technologies used for developing the artificial intelligence module and the web-based healthcare portal. These components ensure efficient implementation, integration, and execution of the system. The software environment consists of programming languages, development frameworks, and supporting tools required for building and deploying the system. Python is utilized for implementing deep learning models, while TensorFlow and Keras are used for model development and training. OpenCV is applied for image preprocessing tasks, and Flask is used to enable communication between the AI module and the web application. For the web portal, HTML, CSS, and JavaScript are used for the frontend, while PHP and MySQL are employed for backend processing and database management. These technologies collectively ensure seamless integration and efficient system performance.

- *Operating System:* Windows 10/11
- *Programming Language:* Python
- *Deep Learning Frameworks:* TensorFlow and Keras
- *Image Processing Library:* OpenCV
- *Dataset:* BraTS (Brain Tumor Segmentation Dataset)
- *Web Framework:* Flask
- *Front-End Technologies:* HTML, CSS, and JavaScript
- *Back-End Technology:* PHP
- *Database Management System:* MySQL
- *Web Server:* WAMP Server
- *Development Tools:* PyCharm

2 LITERATURE SURVEY

2.1 Dataset

Magnetic Resonance Imaging (MRI) is widely regarded as the most reliable and effective imaging modality for detecting brain tumors. It provides high-resolution images with excellent soft tissue contrast, enabling clinicians to visualize intricate brain structures and identify abnormalities with precision. Unlike Computed Tomography (CT) scans, MRI does not use ionizing radiation, making it a safer option for patients and suitable for repeated examinations.

Fig. 2.1 shows that MRI scans consist of multiple modalities, including T1-weighted, T1-contrast enhanced (T1c), T2-weighted, and Fluid Attenuated Inversion Recovery (FLAIR) images. Each modality highlights specific characteristics of brain tissues and tumor structures. For instance, T1-weighted images provide detailed anatomical information, T1c images enhance tumor boundaries, T2-weighted images reveal fluid content, and FLAIR images effectively highlight tumor edema. The combination of these modalities enables accurate diagnosis and precise segmentation of brain tumors.

In the proposed system, the MRI data obtained from the dataset is originally in three-dimensional (3D) volumetric format. To make the data suitable for deep learning models, the 3D MRI volumes are converted into two-dimensional (2D) slices by extracting individual image slices along the axial plane. The dataset used is the BraTS dataset, which contains MRI scans of 369 patients, each consisting of approximately 155 slices, resulting in around 57,000 images. To improve model efficiency, only slices containing tumor regions are selected, reducing the dataset to a few thousand relevant images for training. These slices are then preprocessed using techniques such as normalization and resizing. In this project, the images are resized to a fixed dimension (e.g., 128×128 pixels) to ensure uniform input size for

the models. This preprocessing step reduces computational complexity and improves model performance by providing standardized input data.

Despite its advantages, manual interpretation of MRI scans remains a labor-intensive and subjective process that depends on the expertise of radiologists. The increasing volume of medical imaging data further adds to the workload, leading to diagnostic variability and potential delays. Therefore, automated systems based on artificial intelligence are utilized in the proposed system to assist clinicians in analyzing MRI images efficiently, improving accuracy and reducing dependency on manual interpretation.

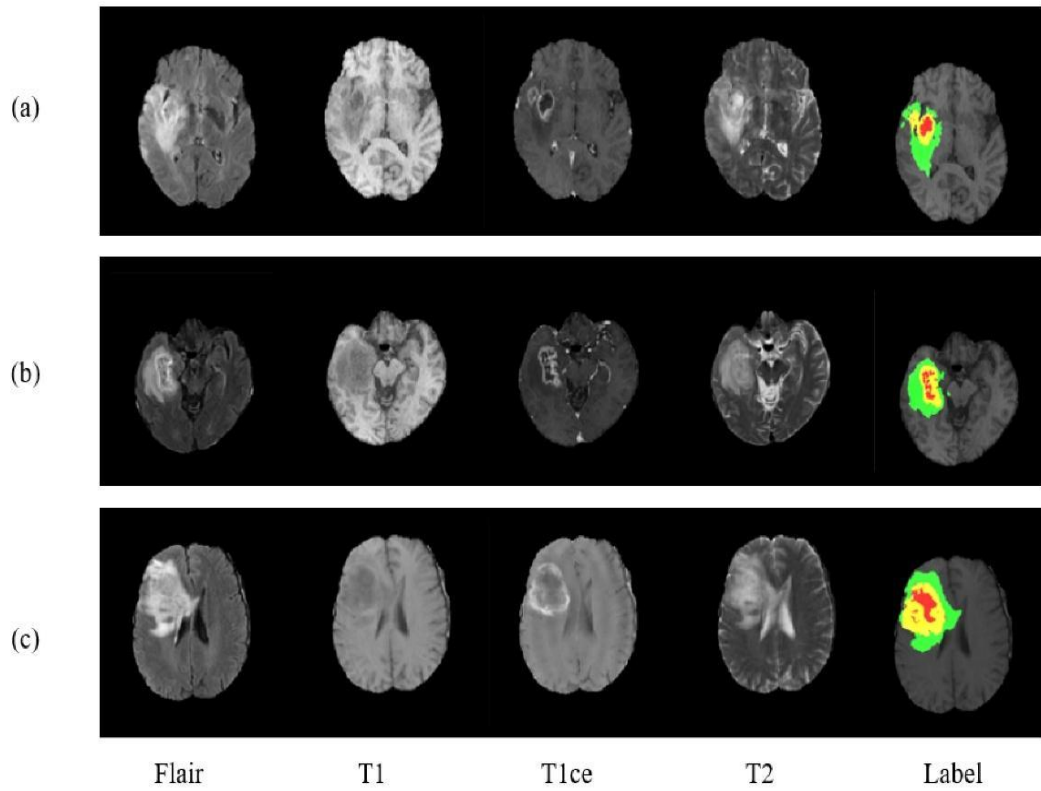


Fig. 2.1 BraTs Dataset

2.2 Convolutional Neural Networks in Medical Imaging

Convolutional Neural Networks (CNNs) are among the most widely used deep learning models for image analysis and computer vision tasks. They are designed to automatically extract hierarchical features from images using convolutional operations, pooling layers, and activation functions. CNNs eliminate the need for manual feature engineering by learning patterns directly from data, making them highly effective for medical image analysis.

In the field of healthcare, CNNs have demonstrated exceptional performance in detecting abnormalities such as tumors, lesions, and fractures. Their ability to capture spatial and contextual information makes them particularly suitable for analyzing MRI scans. By identifying subtle patterns within medical images, CNNs enhance diagnostic accuracy and support clinical decision-making.

In brain tumor detection, CNNs are widely used for identifying abnormal regions and extracting features that distinguish tumor tissues from normal brain structures. In the proposed system, CNN is employed as a feature extraction backbone, enabling accurate analysis of MRI images. These extracted features serve as inputs for advanced segmentation architectures such as U-Net and Attention U-Net.

2.3 U-Net Architecture for Brain Tumor Segmentation

U-Net is a specialized deep learning architecture developed for biomedical image segmentation. It is widely recognized for its effectiveness in accurately segmenting medical images, even with limited training data. The architecture follows an encoder-decoder structure that captures contextual information and reconstructs high-resolution segmentation maps, as illustrated in Fig. 2.2.

The encoder, or contracting path, extracts feature from input images through successive convolution and pooling operations, while the decoder, or expanding path, reconstructs the segmented output. Skip connections between corresponding layers preserve spatial information and enhance segmentation accuracy by combining low-level and high-level features.

U-Net performs pixel-wise classification, enabling precise delineation of tumor boundaries. In brain tumor analysis, it effectively segments tumor regions such as Edema, Tumor Core, and Enhancing Tumor. Due to its robustness and efficiency, U-Net serves as a fundamental component of the proposed system for accurate tumor segmentation.

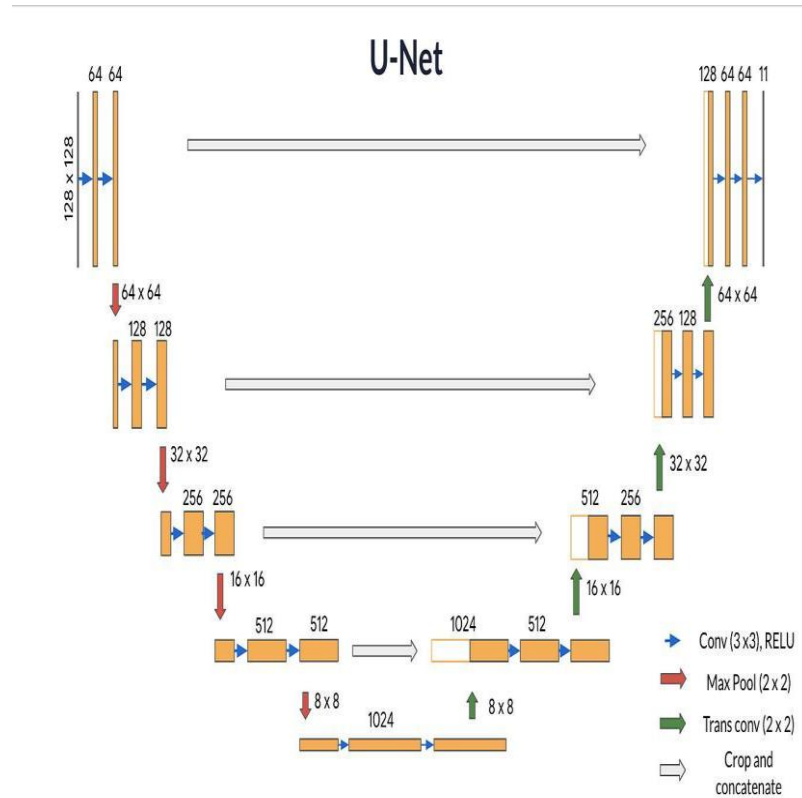


Fig. 2.2 U-Net Architecture

2.4 Attention U-Net for Enhanced Tumor Detection

Attention U-Net is an advanced extension of the traditional U-Net architecture that incorporates attention mechanisms to improve segmentation performance. It introduces attention gates within the network, which allow the model to focus selectively on relevant regions of interest while suppressing unnecessary background information. This enhances the network's ability to capture meaningful features from medical images. By improving feature representation, Attention U-Net increases both the accuracy and efficiency of tumor detection compared to standard segmentation approaches, as illustrated in Fig. 2.3.

The attention mechanism plays a crucial role in medical image analysis, where precise localization of abnormalities is essential. It dynamically highlights important regions in MRI scans, enabling the model to concentrate on tumor-specific features. This reduces false positives and improves the overall segmentation quality. Attention U-Net is particularly effective in identifying small, irregular, and complex tumor regions, including Enhancing Tumor, which are often difficult to detect using conventional methods. Its ability to refine feature selection makes it suitable for handling variations in tumor shape, size, and intensity.

In this project, Attention U-Net is implemented to refine tumor segmentation results obtained from MRI images and improve diagnostic reliability. It works in combination with CNN for feature extraction, while U-Net and Attention U-Net are used for segmentation, forming an efficient detection pipeline. The model enhances the identification of tumor regions such as Edema, Tumor Core, and Enhancing Tumor, ensuring more precise segmentation outputs. Its integration into the system contributes to accurate analysis and supports the development of an automated brain tumor detection system within the web-based healthcare portal.

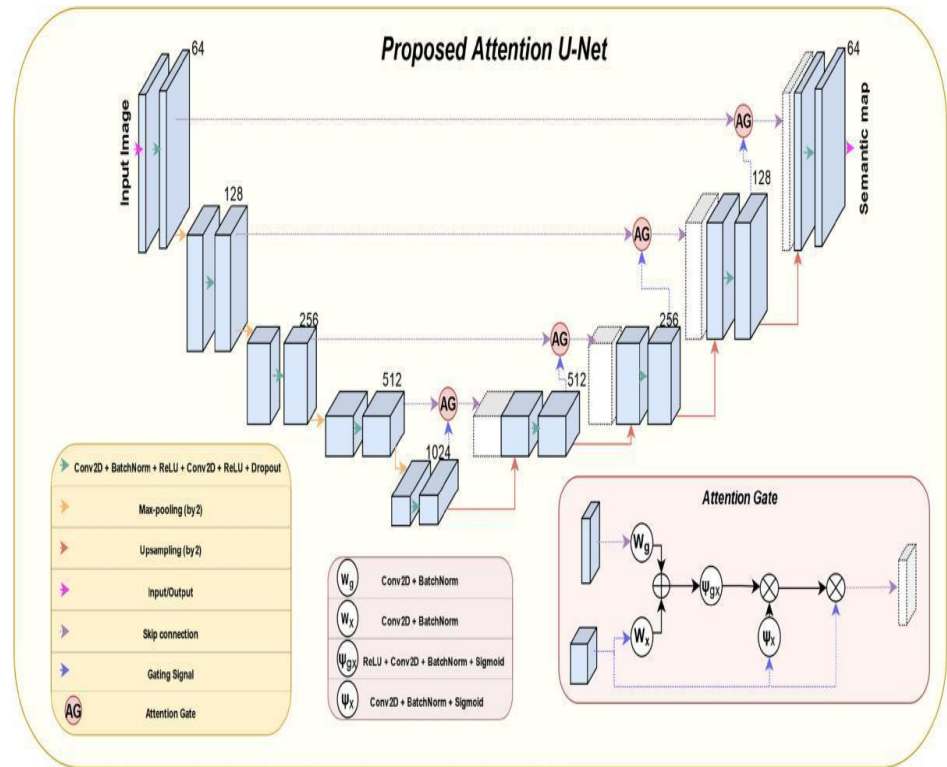


Fig. 2.3 Attention U-Net Architecture

2.5 Web-Based Healthcare Systems

The integration of artificial intelligence with web-based healthcare systems improves the management of medical data and services. Web portals allow structured storage of patient records, diagnostic reports, and appointment details, making information easily accessible. This reduces manual handling of data and supports better organization within the system. The use of web-based platforms also helps in maintaining consistency and accuracy in handling patient-related information.

Web technologies enable interaction between different users such as patients, doctors, and administrators through a single platform. Patients can access their reports and upload medical data, while doctors can review results and manage

patient information. Administrators can monitor system activities and maintain records. This centralized approach reduces delays in communication and helps in managing the workflow more efficiently.

In this project, a Doctor–Patient portal is developed using HTML, CSS, JavaScript, PHP, and MySQL. The artificial intelligence model is connected to the web application using a Flask-based API, which allows MRI images to be processed and results to be returned to the portal. This integration enables users to upload images, view tumor detection results, and manage related data within the system.

In addition, the integration ensures efficient data flow and real-time processing within the system. MRI images uploaded through the portal are securely transmitted to the deep learning model via the Flask API, and the generated results are returned and made accessible through the system and stored efficiently for retrieval.

This seamless interaction between the web interface and the AI module minimizes processing time and improves system responsiveness. Furthermore, security mechanisms such as authentication and role-based access control are implemented to protect sensitive medical information. This integrated approach enhances reliability, supports continuous monitoring, and ensures a smooth and efficient healthcare service delivery.

2.6 Research Gap

Despite significant advancements in brain tumor detection, several limitations persist in existing systems. Many traditional approaches rely on manual feature extraction and lack automation, resulting in reduced accuracy and increased processing time. Furthermore, several deep learning models focus primarily on tumor classification rather than precise segmentation.

Another major limitation of existing research is the absence of integrated healthcare platforms that combine automated diagnosis with patient management systems. Most studies emphasize model development without providing real-time accessibility or centralized data management.

To address these challenges, the proposed system CNN, U-Net, and Attention U-Net were implemented and compared, and based on the analysis, only the *Attention U-Net model was integrated with the web-based Doctor–Patient portal*. By utilizing the BraTS dataset and advanced preprocessing techniques, the system delivers accurate tumor detection, automated reporting, and efficient healthcare management.

The proposed system also emphasizes scalability, usability, and real-time interaction within a clinical environment. By integrating the Attention U-Net model with a user-friendly web interface, the system enables patients to upload MRI scans, doctors to review automated results, and administrators to manage data efficiently through a centralized platform. The incorporation of secure data handling, report generation, and appointment management features further enhances the practical applicability of the system. This holistic approach not only improves diagnostic accuracy but also bridges the gap between artificial intelligence and real-world healthcare services, making the system more reliable and accessible for continuous medical support.

3 SYSTEM ANALYSIS AND DESIGN

3.1 System overview

System Analysis and Design is a critical phase in the Software Development Life Cycle (SDLC) that involves understanding system requirements and transforming them into a structured framework for implementation. This phase ensures that the developed system meets user expectations while maintaining accuracy, reliability, and scalability. It provides a clear understanding of system functionality, operational feasibility, and technical requirements, thereby establishing a strong foundation for successful system development and deployment.

The proposed system, titled *Deep Learning-Based Brain Tumor Detection and Segmentation with an Integrated Doctor–Patient Portal*, is designed to provide an automated, accurate, and efficient solution for brain tumor diagnosis using Magnetic Resonance Imaging (MRI) scans. The system utilizes advanced deep learning models such as Convolutional Neural Networks (CNN), U-Net, and Attention U-Net to detect and segment tumor regions with high precision. Initially, these models were implemented and comparatively evaluated using performance metrics such as Dice Score and Accuracy to determine the most effective model for tumor segmentation. Based on this analysis, Attention U-Net was found to demonstrate superior performance due to its ability to focus on relevant tumor regions and suppress background noise, and is therefore selected as the final model for the system. These models are trained and evaluated using the Brain Tumor Segmentation (BraTS) dataset, ensuring reliable and consistent performance. By automating tumor detection, the system reduces manual workload and enhances diagnostic accuracy.

In addition to the AI-based detection module, the system integrates a web-based Doctor–Patient portal that enables efficient healthcare management. Patients can upload MRI scans and access diagnostic reports, doctors can review segmentation results and provide medical recommendations, and administrators can manage system data and operations. The finalized Attention U-Net model is integrated with the web portal through a Flask-based interface to enable real-time tumor detection and result generation. The system is developed using technologies such as Python, TensorFlow, Keras, OpenCV, Flask, PHP, MySQL, HTML, CSS, and JavaScript. The Flask framework facilitates communication between the AI model and the web portal, enabling real-time prediction and data exchange, while the MySQL database ensures secure storage of patient information and diagnostic results.

Furthermore, the system includes an automated report generation feature, where detailed tumor analysis results, including tumor type, segmentation output, tumor location, and evaluation metrics such as Dice Score and Accuracy, are generated as downloadable PDF reports. Overall, the system provides a comprehensive and scalable solution for intelligent healthcare applications.

3.2 Requirement Analysis

Requirement analysis involves identifying, documenting, and analyzing the needs of users and stakeholders to ensure that the system meets its intended objectives. It serves as a foundation for system design by defining the functionalities and performance expectations of the proposed solution. The Brain Tumor Detection System is designed to support multiple user roles, including patients, doctors, and administrators, each with specific responsibilities and access privileges.

The functional requirements of the system include user authentication, MRI image upload, automated tumor detection, segmentation and classification of tumor

regions, report generation, appointment scheduling, and database management. Patients can register and upload MRI scans, doctors can analyze tumor detection results and provide diagnoses, and administrators can manage users and system data. These functionalities ensure efficient and secure operation of the healthcare portal.

In addition to functional requirements, the system must satisfy several non-functional requirements to ensure quality and reliability. These include high accuracy in tumor detection, efficient system performance, robust data security, user-friendly interfaces, and scalability for future enhancements. The system is designed to handle large volumes of medical data while maintaining speed and precision.

3.3 Feasibility Analysis

Feasibility analysis evaluates the practicality and viability of the proposed system from technical, economic, operational, and scheduling perspectives. This assessment ensures that the system can be successfully developed and implemented within the available resources and constraints.

From a technical standpoint, the system is feasible due to the availability of advanced tools and technologies such as Python, TensorFlow, Keras, OpenCV, Flask, PHP, MySQL, HTML, CSS, and JavaScript. These technologies are widely used in artificial intelligence and web development, ensuring efficient implementation and reliable performance. The use of the BraTS dataset further enhances the system's credibility and accuracy.

Economically, the project is cost-effective as it utilizes open-source software and publicly available datasets. This significantly reduces development and operational expenses, making the system suitable for academic and research purposes. Operational feasibility is ensured through a user-friendly interface that

simplifies interactions for patients, doctors, and administrators. The system is designed to be intuitive and easy to use, thereby facilitating its adoption in healthcare environments.

From a scheduling perspective, the project is feasible within the academic timeline. A structured development approach involving data preprocessing, model training, system integration, testing, and deployment ensures timely completion. This comprehensive feasibility analysis confirms that the proposed system is practical, reliable, and suitable for implementation.

3.4 System Architecture

The system architecture defines the structural framework of the proposed Deep Learning-Based Brain Tumor Detection and Segmentation System integrated with a Doctor–Patient portal. It illustrates the interaction among various hardware and software components required for automated tumor detection and efficient healthcare management. The architecture is designed to ensure high accuracy, scalability, security, and seamless communication between the artificial intelligence module and the web-based healthcare portal. By integrating deep learning techniques with modern web technologies, the system provides a reliable and intelligent solution for medical diagnosis. The overall structure of the proposed system architecture is depicted in Fig. 3.1.

The proposed system follows a multi-tier architecture consisting of the Presentation Layer, Application Layer, Artificial Intelligence Layer, Integration Layer, and Data Layer. This layered approach promotes modularity, maintainability, and flexibility, allowing each component to function independently while interacting effectively with other modules. It simplifies system development, enhances performance, and supports future upgrades without affecting existing functionalities.

3.4.1 Presentation Layer

The Presentation Layer, also known as the user interface layer, is responsible for facilitating interaction between users and the system. It is developed using HTML, CSS, and JavaScript to provide an intuitive and responsive web interface. This layer enables patients to upload MRI scans, doctors to review tumor detection results, and administrators to manage system operations. It includes features such as easy navigation, form validation, and interactive elements to enhance usability.

The interface ensures accessibility and compatibility across multiple devices and browsers, making it suitable for real-world healthcare environments. It also provides real-time feedback to users and improves overall user experience. As shown in Fig. 3.1, this layer acts as the front-end interface connecting users with the backend system.

3.4.2 Application Layer

The Application Layer manages system logic and handles user requests. It is implemented using PHP and operates on a WAMP server environment. This layer is responsible for user authentication, session management, appointment scheduling, report generation, and communication with the database. It ensures secure data processing and efficient handling of system workflows.

In addition, this layer interacts with the Artificial Intelligence module through the Integration Layer using a Flask-based API. When an MRI scan is uploaded, the application layer sends the data to the AI model and receives the processed tumor detection results. It performs input validation, error handling, and secure data transmission, ensuring system reliability. As illustrated in Fig. 3.1, this layer acts as a bridge between the presentation and backend components.

3.4.3 Data Layer

The Data Layer is responsible for storing and managing all system data securely. It is implemented using MySQL, which maintains records such as patient details, doctor information, MRI scans, tumor detection reports, appointments, and administrative data. The database ensures data integrity, consistency, and confidentiality through structured storage and controlled access.

This layer supports efficient data retrieval and storage, enabling real-time system performance. It also allows centralized data management, which improves decision-making in healthcare services. Backup and recovery mechanisms can be incorporated to prevent data loss. As represented in Fig. 3.1, this layer forms the foundation of the system by handling all data-related operations.

3.4.4 Additional System Components and Processing

The Artificial Intelligence Layer serves as the core component of the system, performing automated brain tumor detection and segmentation using advanced deep learning models. It is implemented in Python using TensorFlow, Keras, and OpenCV. Convolutional Neural Networks (CNN) are used for feature extraction, while U-Net and Attention U-Net architectures are used for accurate tumor segmentation. The models are trained using the BraTS dataset to identify tumor regions such as Edema, Tumor Core, and Enhancing Tumor with high precision.

The Integration Layer ensures seamless communication between the web application and the AI module. It is implemented using the Flask framework, which acts as a RESTful API. When an MRI scan is uploaded, the API processes the image using the trained Attention U-Net model and returns the segmentation results to the web portal, enabling real-time interaction and efficient data exchange.

The system also incorporates preprocessing techniques such as image resizing, normalization, and noise reduction to improve model performance and accuracy. Evaluation metrics including Dice Score, Accuracy, Precision, and Recall are used to assess model performance and validate system reliability.

The modular design enhances scalability and maintainability, allowing future enhancements such as cloud deployment, mobile integration, and 3D tumor visualization. The system provides accurate and automated tumor detection, reduces manual workload for healthcare professionals, supports real-time processing, and enables centralized data management. Overall, as shown in Fig. 3.1 the architecture provides a robust, efficient, and intelligent framework for modern healthcare applications.

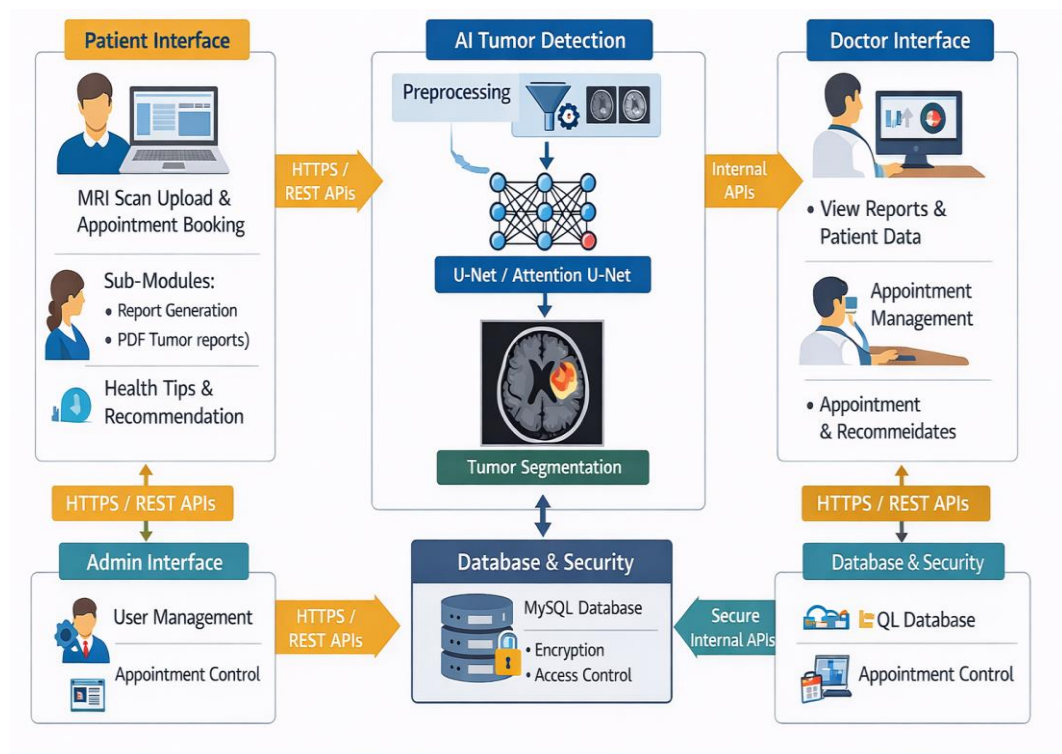


Fig. 3.1 System Architecture

3.5 3D MRI Tumor Detection Architecture

Unlike traditional two-dimensional analysis, three-dimensional imaging captures spatial relationships within brain tissues, enabling improved visualization and precise tumor localization. Fig. 3.2 illustrates the advanced processing of volumetric MRI scans for accurate tumor detection and segmentation. This architecture enhances diagnostic accuracy and supports medical research in neuroimaging.

The process begins with the acquisition of 3D MRI scans, followed by preprocessing steps such as normalization, resizing, and noise reduction. While the Attention U-Net architecture segments tumor regions, including Edema, Tumor Core, and Enhancing Tumor. The segmented results are visualized through the system interface and the final output generated using the Attention U-Net model is included in the downloadable PDF report, enabling easy access and clinical reference. This architecture supports advanced diagnostic capabilities and future integration with augmented reality and telemedicine platforms.

Another significant advantage of the 3D MRI Tumor Detection Architecture is its ability to provide enhanced visualization and accurate estimation of tumor volume. Unlike traditional two-dimensional imaging, three-dimensional analysis captures spatial relationships within brain tissues, enabling precise measurement of tumor size, shape, and location. This capability assists medical professionals in evaluating tumor progression, planning treatment strategies, and monitoring patient recovery. The volumetric representation improves diagnostic confidence and supports advanced clinical decision-making by offering a comprehensive understanding of tumor characteristics.

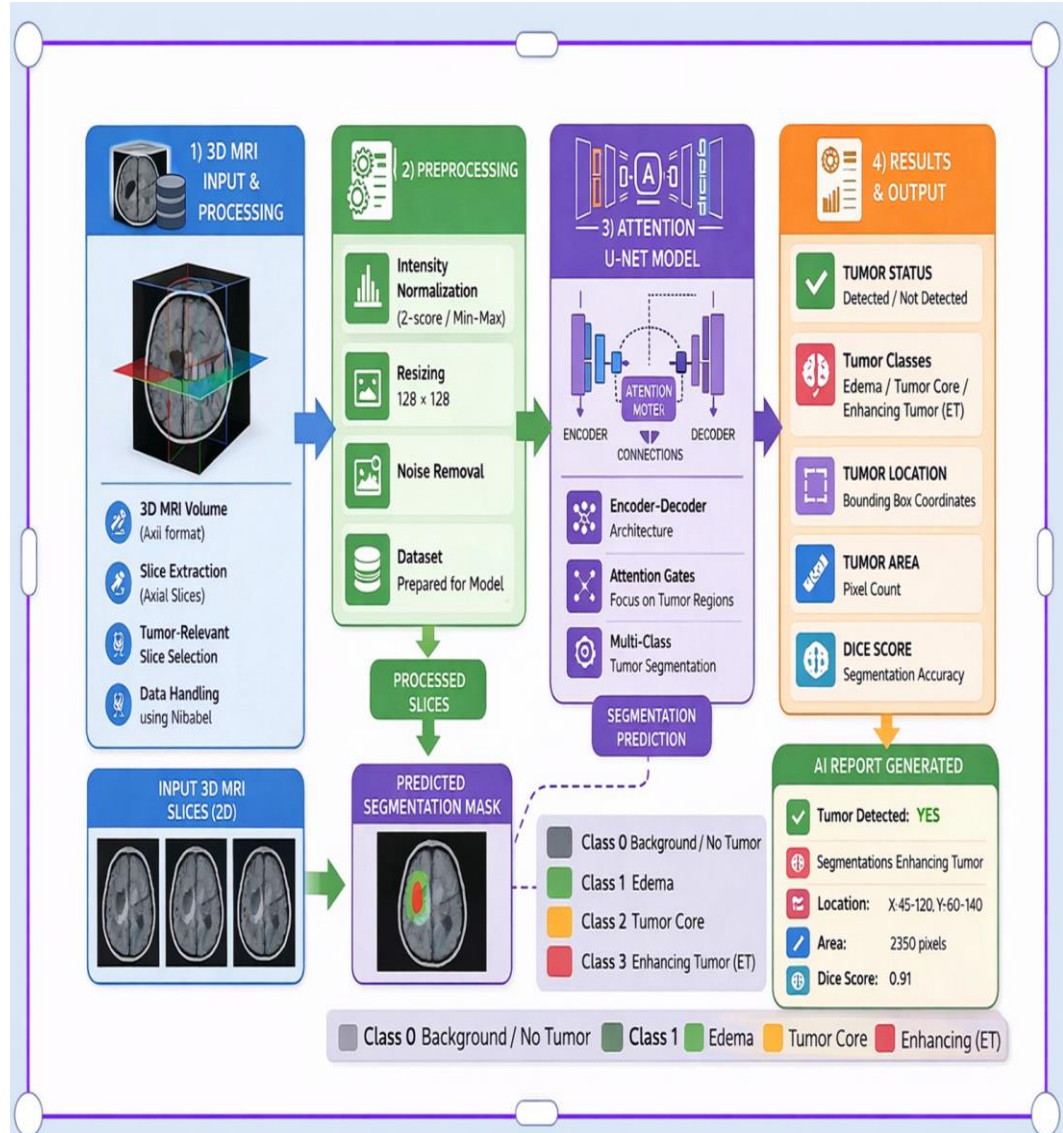


Fig. 3.2 3D MRI Brain Tumor Detection Architecture

3.6 System Workflow

The system workflow describes the step-by-step process involved in detecting and segmenting brain tumors using deep learning techniques. It begins with user registration and authentication, where patients log into the portal and upload MRI

scans for analysis. The uploaded images are transmitted to the Flask API, which forwards them to the trained deep learning model for processing.

The AI model preprocesses the MRI images through normalization and resizing before performing feature extraction and segmentation using the Attention U-Net model, which was selected as the final model based on its improved performance. The model identifies tumor regions and generates prediction outputs, which are then sent back to the web portal. These results are displayed through the system interface and included in downloadable PDF reports for further analysis and clinical reference, while relevant details are managed within the system for access by doctors.

Doctors analyze the AI-generated results and provide medical recommendations, which are subsequently shared with patients through the portal. Administrators oversee system operations, manage user data, and ensure database integrity. This structured workflow ensures accuracy, efficiency, and seamless communication among stakeholders.

Furthermore, the system incorporates result visualization and reporting mechanisms to enhance interpretability and usability. The detected tumor regions are displayed alongside the original MRI images, enabling clear comparison and analysis. Bounding region highlighting and segmentation masks are used to represent tumor localization effectively. Patients are provided with options to view and download diagnostic reports, while doctors can access historical records for continuous monitoring and follow-up analysis. The system also ensures secure data handling through authentication and controlled access, thereby maintaining confidentiality and reliability. This end-to-end workflow supports efficient diagnosis, improves decision-making, and ensures a user-friendly healthcare experience.

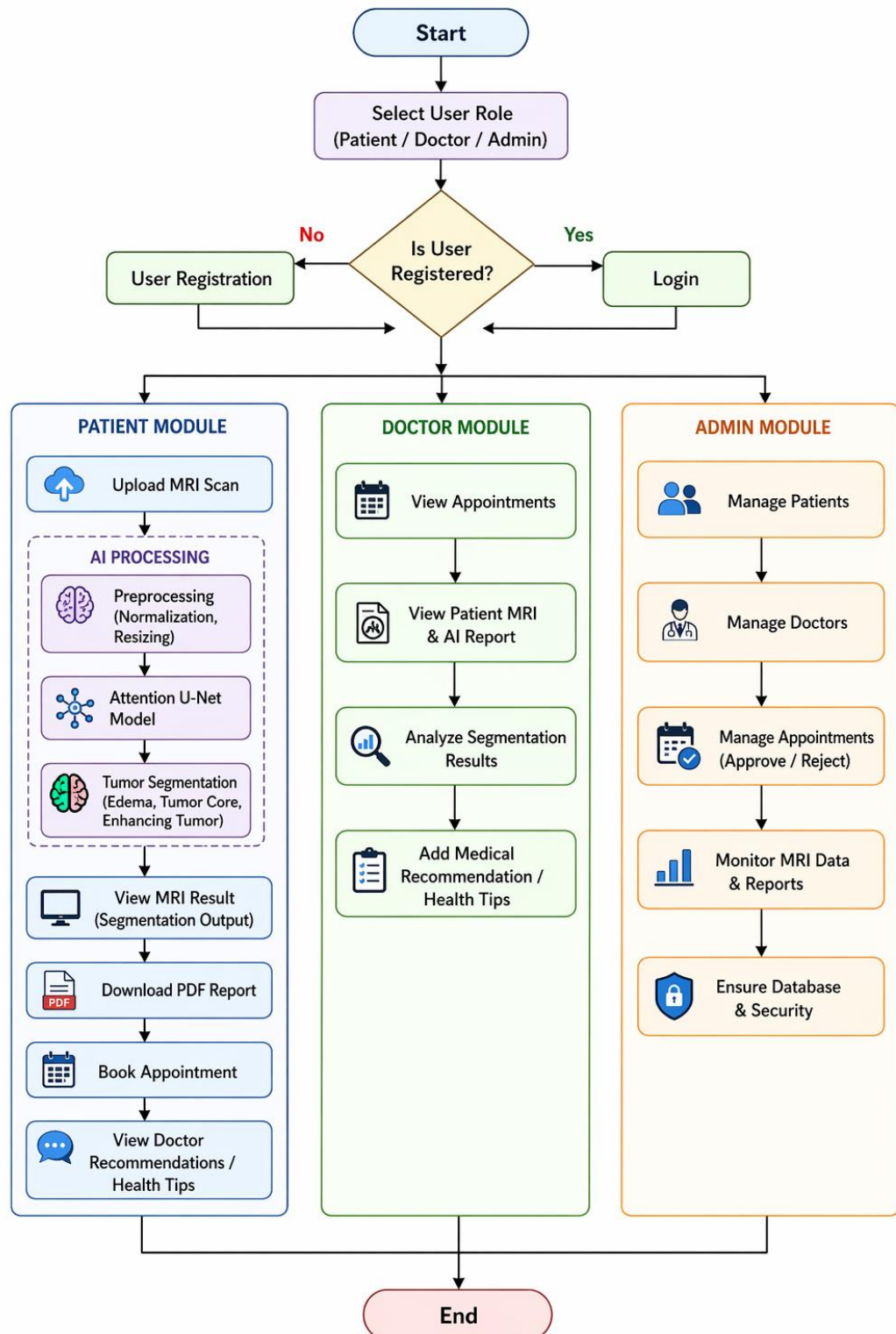


Fig. 3.3 Workflow of the Proposed Brain Tumor Detection System

3.7 DATABASE DESIGN

3.7.1 Database Overview

Database design is a critical component of any information system, as it ensures efficient storage, retrieval, and management of data. In the proposed Deep Learning-Based Brain Tumor Detection and Segmentation System, the database serves as the backbone of the application by maintaining medical records, storing MRI images, managing tumor detection reports, and facilitating communication among patients, doctors, and administrators. A well-structured database enhances system performance, ensures data integrity, and supports secure and reliable healthcare services.

The database is implemented using MySQL, a robust and widely used relational database management system. It is hosted on the WAMP server, enabling seamless integration with PHP and the Flask-based AI module. The relational model ensures structured data storage, eliminates redundancy, and maintains consistency through the use of primary and foreign keys. This chapter presents the database architecture, entity-relationship model, table structures, normalization techniques, and security considerations of the proposed system.

3.7.2 Objectives of Database Design

The primary objective of the database design is to provide an efficient and secure framework for managing healthcare data. The database supports all system modules and ensures smooth interaction between artificial intelligence and the web portal.

- To store and manage patient and doctor information efficiently, structured database tables are created to organize and maintain data systematically. Efficient indexing and relational mapping are

implemented to enable quick access and updates. Accurate and up-to-date information is consistently maintained.

- To maintain records of MRI scans and tumor detection reports, data is securely stored for future reference and analysis. Proper linking mechanisms are established to associate reports with specific patients and their medical history. This supports continuity in diagnosis and effective medical decision-making.
- To ensure data accuracy, integrity, and consistency, validation rules and constraints are enforced within the database. Primary keys and foreign keys are used to prevent data inconsistencies and duplication. Consistent and reliable data is thereby maintained across all system modules.
- To enable secure authentication and authorization, user access is controlled through login credentials and role-based permissions. Authorization mechanisms are implemented to restrict access to sensitive medical data. Confidential information is thereby protected from unauthorized access.
- To facilitate efficient data retrieval and report generation, optimized query processing and indexing techniques are utilized. Reports are dynamically generated based on stored data. This enables quick access to medical insights and supports timely healthcare services.
- To support AI-based tumor detection and segmentation results, outputs generated from deep learning model such as Attention U-Net . These results are properly linked with corresponding MRI

scans and patient records. Seamless integration between the AI module and the web portal is thereby ensured.

- To minimize data redundancy through normalization, data is organized into well-structured tables following normalization rules. Redundant data is eliminated to improve storage efficiency and consistency. This reduces anomalies and enhances database performance.
- To ensure scalability and flexibility for future enhancements, the database is designed with adaptable schema structures. It is capable of handling increasing data volumes and additional system features. This ensures long-term usability and system expansion.
- To provide reliable backup and recovery mechanisms, regular data backups are implemented to prevent data loss. Recovery procedures are established to restore data efficiently in case of system failure. Data availability and system reliability are thereby ensured.
- To enhance interoperability among system modules, standardized data formats and structured relationships are maintained. Seamless data exchange is enabled between patient, doctor, admin, and AI modules. Efficient coordination and integration across the system are thereby achieved.

3.7.3 Database Management System

The proposed system utilizes MySQL as the database management system. MySQL is an open-source relational database that is widely recognized for its reliability, scalability, and high performance. Structured Query Language (SQL) is

supported, enabling efficient data manipulation and retrieval. Seamless integration is achieved with PHP and Flask, making it a suitable choice for web-based healthcare applications.

- To ensure high performance and scalability, MySQL is designed to handle large volumes of data efficiently. It supports optimized query execution and indexing mechanisms. This enables the system to perform reliably even as data size and user load increase.
- To provide secure and reliable data storage, MySQL incorporates advanced security features and data protection mechanisms. Data is stored in a structured and controlled environment. This ensures the safety and reliability of sensitive healthcare information.
- To support relational database structures, MySQL organizes data into tables with well-defined relationships. Primary keys and foreign keys are used to maintain data integrity. This structured approach ensures efficient data organization and consistency.
- To enable efficient query execution using SQL, MySQL supports powerful query processing capabilities. Complex queries can be executed quickly using optimized algorithms. This improves system performance and allows faster data retrieval.
- To ensure compatibility with PHP and Python, MySQL is designed to integrate seamlessly with backend technologies. It works effectively with PHP for web development and Python for AI integration. This enables smooth communication between different system components.

3.7.4 User Table

Table 3.1 describes the *Users Table*, which serves as the central repository for authentication and authorization in the system. It stores the login credentials of all individuals who access the platform, including administrators, doctors, and patients. This table plays a crucial role in maintaining secure access and ensuring efficient management of system users. Each user is assigned a unique identifier (*user_id*) to guarantee distinct identification and data integrity.

The username attribute is used for authentication, while the password is securely stored in encrypted form to protect sensitive information. role attribute defines the level of access assigned to each Admin, Doctor, or Patient, thereby enabling effective role-based access control. Additionally, the *created_at* field records the date and time of account creation for auditing and monitoring purposes.

Table 3.1 User Table

Field Name	Data Type	Description
user_id	INT (PK)	Unique user identifier
Username	VARCHAR (50)	Login username
Password	VARCHAR (25)	Encrypted password
Role	VARCHAR (20)	User role (Admin, Doctor, Patient)
created_at	TIMESTAMP	Account creation date

3.7.5 Patients Table

Table 3.2 presents the *Patients Table*, which stores detailed information about patients registered in the system. It contains essential demographic and medical data necessary for diagnosis and treatment. The table includes attributes such as patient name, age, gender, contact details, and medical history. This information enables patients to upload MRI scans, access tumor detection reports, and schedule appointments with doctors.

Table 3.2 Patients Table

	Data Type	Description
patient_id	INT (PK)	Unique patient identifier
Name	VARCHAR (20)	Patient name
Age	INT	Age of the patient
Gender	VARCHAR (10)	Gender
Phone	VARCHAR (10)	Contact number
Email	VARCHAR (20)	Email address

3.7.6 Doctors Table

Table 3.3 illustrates the *Doctors Table*, which maintains information about medical professionals using the system. Each doctor is assigned a unique `doctor_id` linked to the Users Table for authentication and authorization. The table includes attributes such as name, specialization, contact information, and professional credentials. It enables doctors to review MRI scans, analyze tumor detection results, provide medical recommendations, and manage patient appointments. This table ensures effective communication between patients and healthcare providers, thereby enhancing the quality of medical services.

Table 3.3 Doctors Table

Field Name	Data Type	Description
doctor_id	INT (PK)	Unique doctor identifier
Name	VARCHAR (20)	Doctor name
Specialization	VARCHAR (30)	Medical specialization
Phone	VARCHAR (10)	Contact number
Email	VARCHAR (20)	Email address
Experience	INT	Years of experience

3.7.7 Admin Table

Table 3.4 describes the *Admin Table*, which contains information related to system administrators responsible for managing and supervising the platform. Each administrator is identified by a unique `admin_id`, associated with the Users Table. The table includes details such as administrator name, contact information, and system privileges. Administrators oversee user management, maintain databases, monitor system activities, and ensure data security and integrity.

Additionally, the Admin Table stores login credentials to ensure secure authentication of administrators. It supports role-based access control, allowing different levels of permissions for system management. The table also maintains records of administrative activities for monitoring and auditing purposes. This helps in enhancing system security, accountability, and efficient management of platform operations.

Table 3.4 Admin Table

Field Name	Data Type	Description
admin_id	INT (PK)	Unique administrator ID
username	VARCHAR (20)	Admin username
password	VARCHAR (25)	Encrypted password
email	VARCHAR (20)	Email address

3.7.8 Tumor_Reports Table

Table 3.5 illustrates the *Tumor Reports Table*, which stores the results generated by the artificial intelligence-based tumor detection module. Each report is uniquely identified by a report_id and is linked to the corresponding patient and MRI scan. The table includes information such as tumor type, segmentation results, prediction accuracy, detection date, and generated report files. It enables doctors and patients to review diagnostic outcomes and supports clinical decision-making. This table plays a crucial role in providing reliable and automated medical analysis and Details Stored.

Table 3.5 Tumor_Reports Table

Field Name	Data Type	Description
report_id	INT (PK)	Unique report identifier
patient_id	INT (FK)	References Patients table
scan_id	INT (FK)	References MRI_Scans table
tumor_type	VARCHAR (30)	Type of tumor detected
segmentation_result	VARCHAR (30)	Path of segmented image
dice_score	FLOAT	Model performance metric
detection_date	DATE	Date of detection

3.7.9 Appointments Table

Table 3.6 describes the *Appointments Table*, which manages scheduling and coordination between patients and doctors. Each appointment is assigned a unique `appointment_id` and is linked to both patient and doctor records. The table includes attributes such as appointment date, time, consultation status, and remarks. It facilitates efficient appointment booking, tracking, and management, ensuring smooth communication and timely medical consultation.

Table 3.6 Appointments Table

Field Name	Data Type	Description
appointment_id	INT (PK)	Unique appointment identifier
patient_id	INT (FK)	References Patients table
doctor_id	INT (FK)	References Doctors table
appointment_date	DATE	Scheduled appointment date
Status	VARCHAR(50)	Appointment status
Remarks	TEXT	Doctor's remarks

3.7.10 Health Tips Table

Table 3.7 presents the *Health Tips Table*, which stores informative healthcare guidance provided by doctors and administrators. Each health tip is identified by a unique `tip_id` and includes attributes such as title, description, publication date, and author details. This table enables users to access valuable information related to brain health, tumor awareness, preventive measures, and post-treatment care. It enhances patient education and promotes overall well-being.

Table 3.7 Health Tips Table

Field Name	Data Type	Description
tip_id	INT (PK)	Unique tip identifier
Title	VARCHAR (150)	Title of the health tip
Description	TEXT	Tip content
published_date	DATE	Publication date

4 MODULE DESIGN

4.1 Module Overview

Module design is a crucial phase in the system development process that involves decomposing a complex system into smaller, manageable, and independent components known as modules. This structured approach enhances system efficiency, improves maintainability, ensures scalability, and simplifies debugging and testing. Each module performs a specific function while interacting seamlessly with other modules to achieve the overall objectives of the system. A well-defined modular architecture ensures reliability, flexibility, and effective integration of hardware and software components.

The proposed Deep Learning-Based Brain Tumor Detection and Segmentation System with an Integrated Doctor–Patient Portal is designed using a modular approach to automate tumor detection, facilitate medical diagnosis, and manage healthcare data securely. The system integrates artificial intelligence with web-based technologies, enabling accurate analysis of MRI scans and efficient communication among patients, doctors, and administrators. The modular structure ensures smooth data flow, enhances performance, and supports future enhancements such as cloud integration and mobile accessibility.

The major modules of the proposed system include the Patient Module, Doctor Module, Admin Module, Detection Module, and Database Module. Each module is described in detail in the following sections with its objectives, functionalities, inputs, outputs, and technologies used.

4.2 Patient Module

The Patient Module serves as the primary interface between patients and the system, enabling access to diagnostic services through a secure and user-friendly web portal. The objective of this module is to provide an efficient platform through which MRI brain scans can be uploaded, tumor detection results can be viewed, and communication with healthcare professionals can be facilitated. Secure data handling, ease of access, and transparency in healthcare services are ensured within this module. Once the MRI scan is uploaded by the patient, it is transmitted to the detection module through the Flask API, where automated analysis is performed. The generated results are then processed and displayed through the system interface and included in downloadable PDF reports on the patient's dashboard for review.

The objectives of the Patient Module are defined to ensure effective system usage and improved healthcare accessibility. A secure platform is provided for MRI scan submission, ensuring that sensitive medical data is safely transmitted and stored. Patients are enabled to access diagnostic reports online, allowing convenient and timely review of medical results. Communication with healthcare professionals is facilitated through the system to support better consultation and guidance. Accessibility and efficiency in medical services are enhanced by providing a centralized and easy-to-use interface for patients.

The Patient Module incorporates several key functionalities to support patient interaction and system operations. Patient registration and secure login authentication are implemented to ensure authorized access. MRI brain scan images are uploaded through the interface for analysis by the detection module. Tumor detection and segmentation results are displayed for patient review, and diagnostic reports can be downloaded for further reference. Appointment booking and management functionalities are provided to enable interaction with doctors. Medical history and previously generated reports are made accessible to patients. Health tips and recommendations are provided to promote awareness and

preventive care. Additionally, personal profile details can be updated, and notifications and system updates are provided to users to keep patients informed about system activities.

The inputs required for this module include patient personal information, login credentials, and MRI brain scan images, which are submitted through the web interface. The outputs generated from this module include segmented tumor images, diagnostic reports, appointment confirmations, and medical recommendations, which are displayed to the patient after processing.

The technologies used for implementing the Patient Module include HTML, CSS, and JavaScript for the frontend interface, PHP for backend processing and server-side logic, and MySQL for database management. These technologies are integrated to ensure efficient system performance, secure data handling, and seamless interaction between the user interface and backend components.

4.3 Doctor Module

The Doctor Module is designed to assist medical professionals in reviewing patient data and analyzing AI-generated tumor detection results. Through this module, clinical decision-making is enhanced by providing accurate and reliable diagnostic information. Doctors are enabled to access uploaded MRI scans, validate segmentation outputs generated by the artificial intelligence model, and provide appropriate treatment recommendations. By integrating artificial intelligence with medical expertise, reliable and efficient healthcare services are ensured within the system.

The objectives of the Doctor Module are defined to support effective diagnosis and patient management. Doctors are supported in accurate diagnosis and treatment planning through access to structured patient data and AI-generated

insights. Tumor detection results produced by the artificial intelligence module are made accessible for professional evaluation and validation by doctors.

The Doctor Module includes several key functionalities to support medical operations. Secure login and authentication mechanisms are implemented to ensure authorized access for doctors. Patient profiles and medical histories are made available for review to support informed decision-making. Uploaded MRI scans and segmentation outputs are accessed and analyzed within the system. AI-generated tumor detection reports are reviewed and validated by medical professionals. Diagnoses and treatment recommendations are provided based on the analysis of results. Patient consultations and appointments are managed efficiently through the system. Medical records are updated regularly to reflect patient progress, and continuous monitoring along with follow-up treatments is facilitated to ensure effective healthcare delivery.

The inputs required for this module include patient data, MRI scans, and AI-generated tumor detection results, which are provided through the system database and detection module. The outputs generated include the doctor's diagnosis and treatment recommendations, updated patient medical records, and consultation reports, which are stored in the database and made accessible through the portal.

The technologies used for implementing the Doctor Module include HTML, CSS, and JavaScript for the frontend interface, PHP for backend processing and system logic, and MySQL for database management. These technologies are integrated to ensure secure data handling, efficient processing, and seamless interaction between system components.

4.4 Admin Module

The Admin Module is designed to oversee and manage the overall functioning of the system. Through this module, smooth operation, security, and reliability of the healthcare portal are ensured. Administrative control is provided over user accounts, system data, and application activities, enabling effective system management. This module plays a vital role in maintaining data integrity, monitoring system performance, and ensuring compliance with required security standards.

The objectives of the Admin Module are defined to ensure efficient system administration and control. System users and resources are managed efficiently through centralized administrative functions. Data integrity and system security are ensured by implementing appropriate control mechanisms and monitoring strategies. Overall system performance is continuously monitored and maintained to ensure reliable and uninterrupted operation of the healthcare portal.

The Admin Module includes several key functionalities to support administrative tasks. Secure administrator login and authentication mechanisms are implemented to ensure authorized access. Patient and doctor registrations are managed, and user accounts are added, updated, or deleted as required. MRI uploads and tumor detection reports are monitored to ensure proper system functioning. Appointments and system records are managed efficiently through the portal. System and user reports are generated to support analysis and decision-making. Health tips and notifications are updated to keep users informed. Database integrity is maintained through regular monitoring and backup mechanisms. System performance and activities are continuously monitored to ensure stability and efficiency.

The inputs required for this module include user data, administrative commands, and system records, which are processed to perform various

administrative operations. The outputs generated include user management reports, system performance summaries, and updated healthcare information, which are made accessible through the system for monitoring and management purposes.

4.5 Detection Module

The Detection Module is designed as the core component of the proposed system, responsible for automated brain tumor detection and segmentation. Advanced deep learning algorithms are utilized to analyze MRI images with high accuracy and efficiency. Through this module, manual workload is significantly reduced while diagnostic precision is enhanced. The models are trained using the BraTS dataset, and seamless integration with the web portal is achieved through a Flask API to enable real-time predictions and analysis.

The objectives of the Detection Module are defined to ensure intelligent and automated medical analysis. Brain tumor detection and segmentation are automated using deep learning techniques to minimize human intervention. Diagnostic accuracy is improved by applying advanced neural network models capable of extracting meaningful features from MRI images. Reliable and real-time medical analysis is provided through efficient model execution and integration with the web-based system.

The Detection Module incorporates several key functionalities to perform image processing and tumor analysis. MRI images are preprocessed through resizing and normalization to ensure uniform input for the models. Noise reduction and image enhancement techniques are applied to improve image quality. Feature extraction is performed using Convolutional Neural Networks (CNN), followed by tumor segmentation using the U-Net architecture. Enhanced detection performance is achieved using Attention U-Net, which focuses on relevant tumor regions. Tumor regions such as Edema, Tumor Core, and Enhancing Tumor are identified accurately. Segmentation masks and prediction outputs are generated and stored.

Model performance is evaluated using metrics such as Dice Score, Accuracy, Precision, and Recall to ensure reliability.

The inputs required for this module include MRI brain scan images uploaded by patients and the BraTS dataset used for model training and validation. The outputs generated include segmented tumor images, tumor classification results, and detailed diagnostic and analytical reports, which are provided to the system for further use.

The technologies used for implementing the Detection Module include Python as the programming language, with libraries such as TensorFlow, Keras, OpenCV, and NumPy for model development and image processing. The Flask framework is used for API integration, and the BraTS dataset is utilized for training and validating the deep learning models. These technologies are integrated to ensure accurate, efficient, and real-time tumor detection within the system.

4.6 Database Module

The Database Module is designed to store, manage, and retrieve system data in a secure and efficient manner. Data consistency, confidentiality, and integrity are ensured through structured storage, enabling seamless communication among system modules. The database serves as the backbone of the application by organizing all system-related information in a well-defined format, thereby supporting reliable performance and smooth operation.

The objectives of this module include maintaining secure storage of sensitive medical data and enabling efficient data retrieval and management. Patient, doctor, and administrator details are stored in structured tables, while MRI scan records and tumor detection results are systematically managed and linked with corresponding patient information.

5 USER INTERFACE DESIGN

5.1 Introduction

The User Interface (UI) Design plays a vital role in ensuring seamless interaction between users and the proposed Brain Tumor Detection and Segmentation System. A well-designed interface enhances usability, accessibility, and efficiency by enabling users to interact with the system intuitively. The UI of this project is designed to support three primary stakeholders—patients, doctors, and administrators—through a secure and user-friendly web portal.

The interface has been developed using modern web technologies such as HTML, CSS, JavaScript, and PHP, ensuring responsiveness and compatibility across multiple devices and browsers. The design emphasizes clarity, simplicity, and accessibility, allowing users to perform tasks such as uploading MRI scans, viewing reports, managing appointments, and monitoring system activities efficiently. This chapter presents the design and implementation of various user interfaces developed for the system. Each interface is illustrated with screenshots to provide a visual understanding of the system's functionality and user interaction.

5.2 Design Objectives of the User Interface

The design objectives of the system are defined to ensure the development of a user-friendly and efficient interface that supports smooth interaction among all users. An intuitive and easy-to-use interface is created to enhance user experience, particularly for non-technical users. Easy navigation across different system modules is ensured so that users can access required functionalities without difficulty. Secure login and authentication mechanisms are implemented to protect user data and restrict unauthorized access.

Efficient handling of MRI image upload and analysis is enabled through well-designed interface components. Tumor detection results are displayed in a clear and accurate manner to support better understanding and interpretation by users. Responsive and visually appealing layouts are designed to improve usability and engagement. Data confidentiality and integrity are maintained through secure design practices, ensuring that sensitive healthcare information is protected at all stages.

Seamless interaction among patients, doctors, and administrators is facilitated through a well-structured interface design. The system is developed to enhance accessibility and usability, allowing users to interact with the application efficiently regardless of their technical background. Overall, the design objectives are achieved by ensuring a balance between functionality, security, and user experience.

5.3 Design Principles of the User Interface

The design of the user interface is developed based on standard Human-Computer Interaction (HCI) principles to enhance usability, efficiency, and overall user experience. These principles are applied to ensure consistency, clarity, and accessibility throughout the system. A simple and clean layout is maintained with minimal complexity so that users can interact with the system easily. Consistency is ensured through the use of uniform fonts, color schemes, and navigation patterns across all modules, which improves familiarity and reduces user confusion.

Responsiveness is incorporated into the design to ensure compatibility across various devices, including desktops, tablets, and mobile platforms. Accessibility is emphasized by providing easy-to-read text, intuitive controls, and well-structured interfaces, enabling users of different technical backgrounds to interact with the system effectively. Security considerations are integrated through role-based access control and secure authentication mechanisms to protect sensitive healthcare data.

A user-centric design approach is followed, where interfaces are tailored according to the specific roles of patients, doctors, and administrators. Efficiency is achieved by minimizing user effort through streamlined workflows and optimized interaction processes. Clarity is ensured by presenting tumor detection results and diagnostic reports in a clear and understandable format, thereby supporting accurate interpretation and effective decision-making.

5.4 User Interface Development Tools

The user interface of the proposed system is developed using industry-standard tools and technologies to ensure high performance, scalability, and responsiveness. Each tool is selected based on its capability to support efficient development, seamless integration, and user-friendly interaction within the healthcare application.

HTML5 is used to define the structure of the web pages in the system. All interface components such as forms, input fields, buttons, and navigation elements are created using HTML. It provides the foundational layout for modules such as patient registration, MRI upload, doctor dashboards, and admin panels. Semantic elements are utilized to improve readability and maintainability of the code.

CSS3 is applied to enhance the visual appearance and layout of the web pages. Styling features such as colors, fonts, spacing, alignment, and responsive design elements are implemented using CSS. It ensures that the interface is visually appealing and consistent across different modules. CSS is also used to design structured layouts for dashboards and reports.

Bootstrap is utilized to develop a responsive web design that adapts to different screen sizes and devices. Predefined classes and grid systems are used to create flexible layouts for desktops, tablets, and mobile devices. It ensures uniform

styling and reduces development time by providing ready-to-use components such as navigation bars, buttons, and forms.

PHP is used as the server-side scripting language to handle backend processing. It manages user authentication, session handling, form processing, and communication with the database. PHP scripts are used to process data submitted from the frontend and to generate dynamic content for different user roles such as patients, doctors, and administrators.

MySQL is used as the database management system to store and manage all system data. It is integrated with PHP to perform operations such as data insertion, retrieval, updating, and deletion. Patient records, MRI scans, tumor detection results, appointments, and user details are stored in structured tables within the database.

WAMP Server is used as the local server environment for hosting and testing the web application. It provides an integrated platform that includes Apache server, MySQL database, and PHP support. The system is deployed locally using WAMP during development to ensure smooth execution and testing of all modules.

PyCharm are used as development environments for coding and testing the application. PyCharm is used for developing and testing the Python-based AI module. Flask API is used to integrate the artificial intelligence model with the web application. It acts as a communication bridge between the frontend and the AI module developed in Python. When an MRI image is uploaded, it is sent to the Flask API, where it is processed by the trained model, and the prediction results are returned to the web portal in real time.

5.5 User Interface Modules

5.5.1 Home Page

The Home Page serves as the primary gateway to the Brain Tumor Detection System. It provides users with an overview of the system's objectives, features, and technological advancements. The page introduces the importance of early brain tumor detection using artificial intelligence and guides users toward secure login and registration options. Fig. 5.1 illustrates the Home Page of the Brain Tumor Detection System. The layout is designed to be visually appealing and informative, incorporating clear navigation menus, descriptive content, and responsive design elements. It establishes the credibility of the system and ensures engaging user experience for all stakeholders.

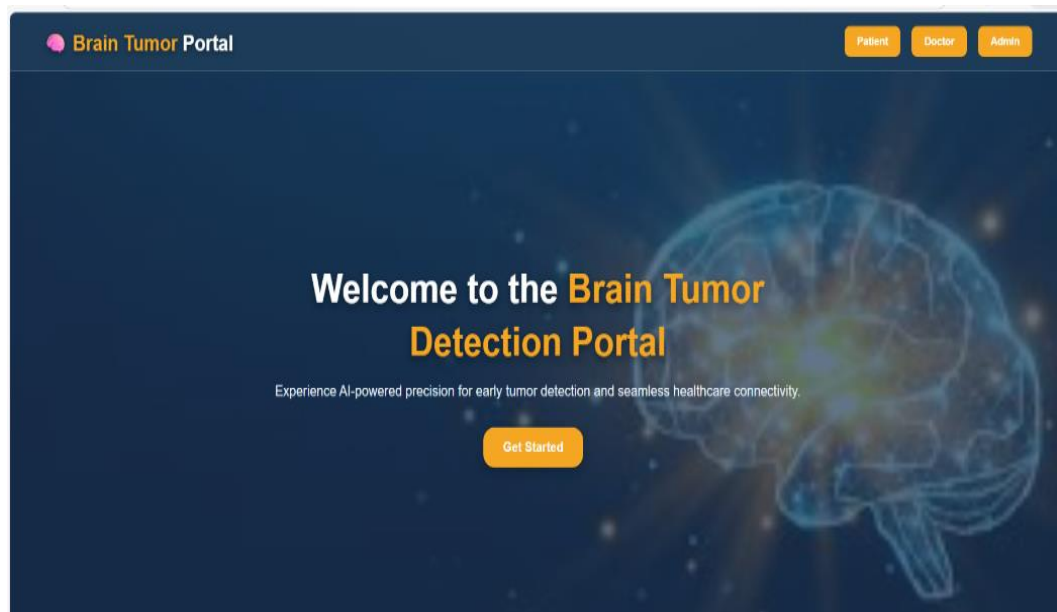
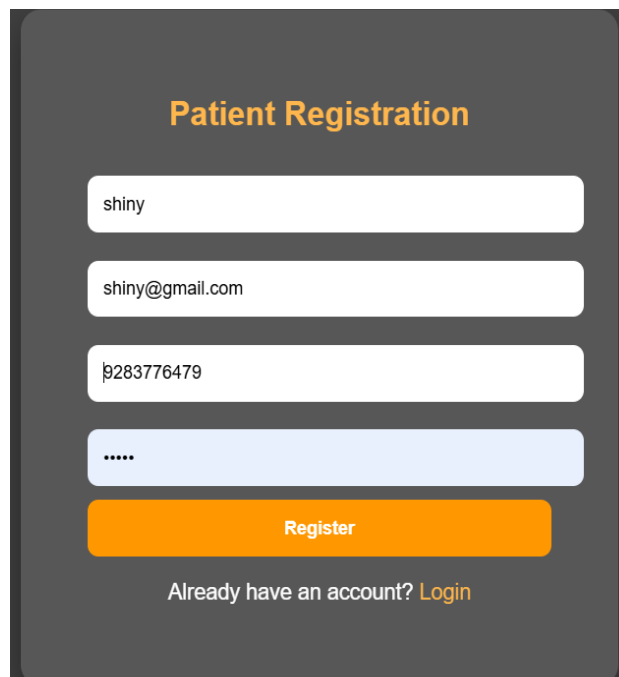


Fig. 5.1 Home Page of the Brain Tumor Detection System

5.5.2 Patient Registration Page

The Patient Registration Page enables new users to create accounts by providing essential personal and medical details. The data entered is securely stored in the MySQL database for future reference and diagnosis. Fig. 5.2 illustrates the Patient Registration Page. The interface is designed for ease of use, ensuring accurate data entry through validation mechanisms. Successful registration grants patients access to system services such as MRI uploads and report viewing.

Furthermore, input validation is implemented to ensure that all required fields are correctly filled and invalid data entries are prevented. Error messages are displayed to guide users in providing accurate information. This enhances data reliability and improves the overall user experience during the registration process.

The image shows a digital form titled "Patient Registration" in orange text. Below the title are four input fields: the first contains "shiny", the second contains "shiny@gmail.com", the third contains "p283776479", and the fourth is a password field with five dots. Below these fields is an orange "Register" button. At the bottom, there is a link that says "Already have an account? Login".

Patient Registration

shiny

shiny@gmail.com

p283776479

.....

Register

Already have an account? [Login](#)

Fig. 5.2 Patient Registration Page

5.5.3 User Login Page

The User Login Page provides secure authentication for patients, doctors, and administrators. It ensures that only authorized users can access the system, thereby maintaining data privacy and security. Role-based access control allows users to interact with modules specific to their responsibilities. Fig. 5.3 shows the User Login Page. The interface includes validation mechanisms and encrypted password authentication to enhance reliability. Its simple and intuitive design ensures quick and secure login.

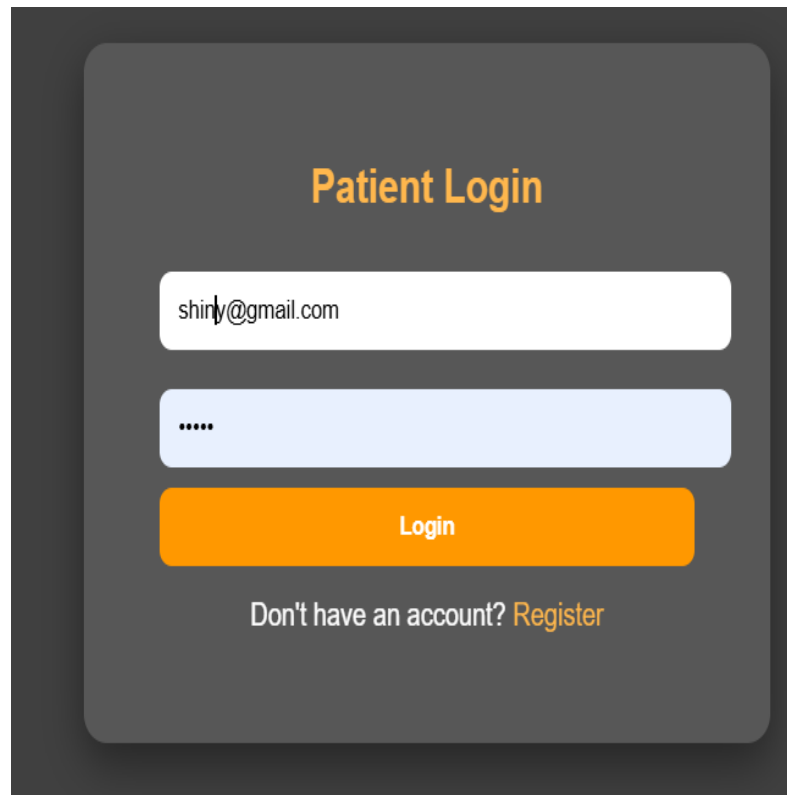
The image shows a 'Patient Login' form. At the top, the title 'Patient Login' is displayed in orange. Below it, there are two input fields: the first is for an email address, containing 'shirly@gmail.com', and the second is for a password, represented by five dots. An orange 'Login' button is positioned below the password field. At the bottom, there is a link that says 'Don't have an account? Register', with 'Register' in orange.

Fig. 5.3 User Login Page

5.5.4 Patient Dashboard

The Patient Dashboard serves as the central interface for patients, providing access to essential services such as MRI upload, report viewing, and appointment scheduling. It enables patients to monitor their health records conveniently. Fig. 5.4 illustrates the Patient Dashboard. The dashboard is designed with intuitive navigation and organized sections, ensuring ease of use and accessibility. It empowers patients with real-time diagnostic insights and secure access to their medical information.

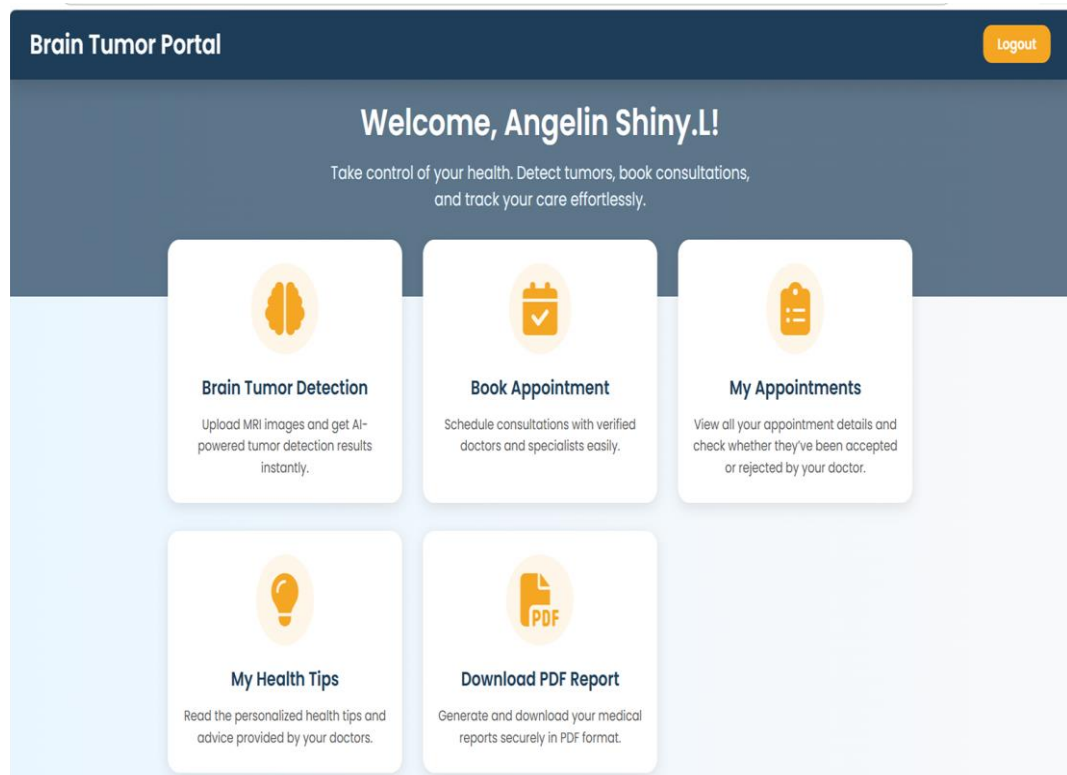


Fig. 5.4 Patient Dashboard

5.5.5 MRI Image Upload Interface

The MRI Image Upload Interface allows patients to submit brain MRI scans for tumor detection and acts as a bridge between the user and the artificial intelligence system. The uploaded images are transmitted to the backend through the Flask API, where they are processed by the Attention U-Net model for analysis. Fig. 5.5 illustrates the MRI Image Upload Interface. The design of the interface ensures smooth interaction and supports efficient data flow between the user and the system.

Furthermore, the interface incorporates validation mechanisms to ensure that only supported file formats and appropriate image sizes are uploaded. Secure transmission protocols are followed to protect sensitive medical data during the upload process. Clear instructions and feedback messages are provided to guide users, improving usability and ensuring reliable operation of the system.

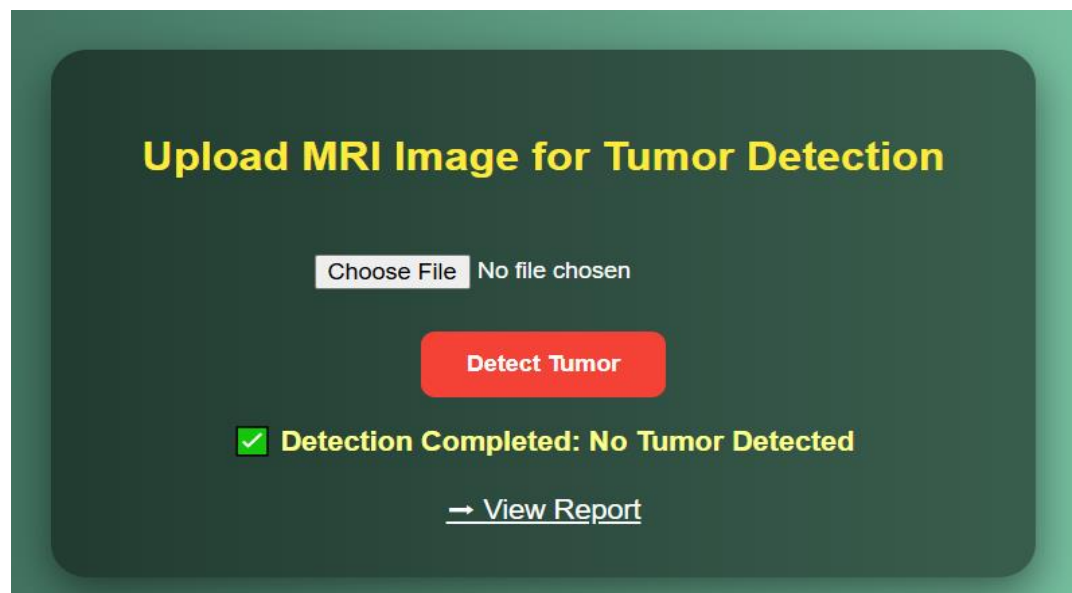


Fig. 5.5 MRI Image Upload Interface

5.5.6 Tumor Detection Result Page

The Tumor Detection Result Page displays outputs generated by deep learning models. It presents the original MRI image alongside the segmented tumor region, enabling accurate visualization and interpretation. Fig. 5.6 illustrates the Tumor Detection Result. This interface provides essential diagnostic insights and assists doctors in clinical decision-making. Patients can also download results for further consultation.

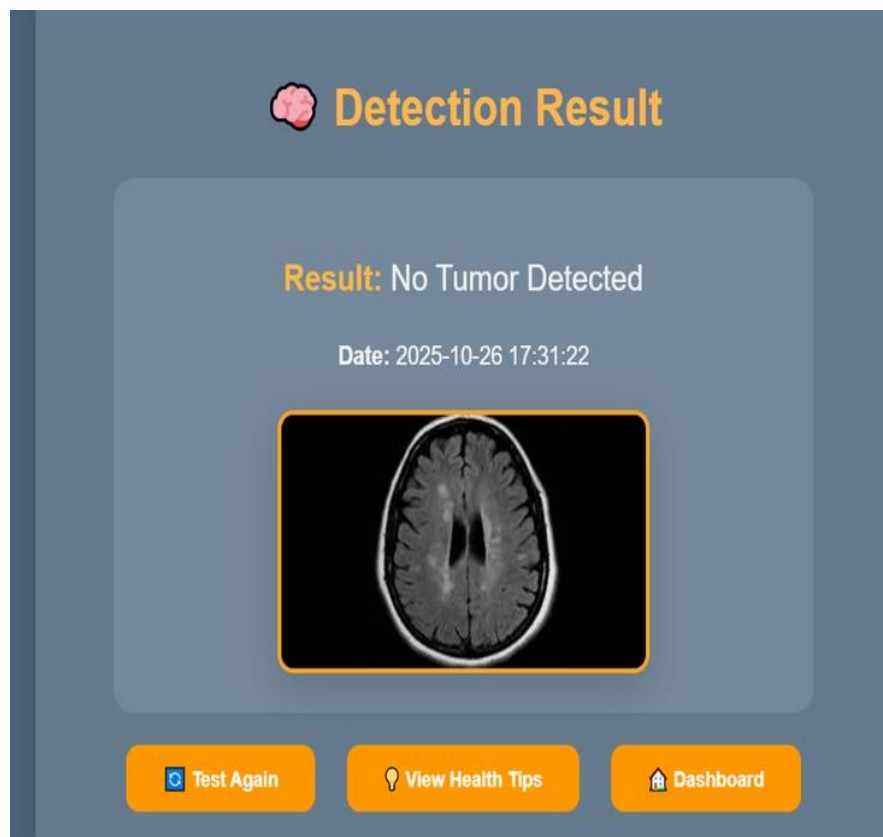
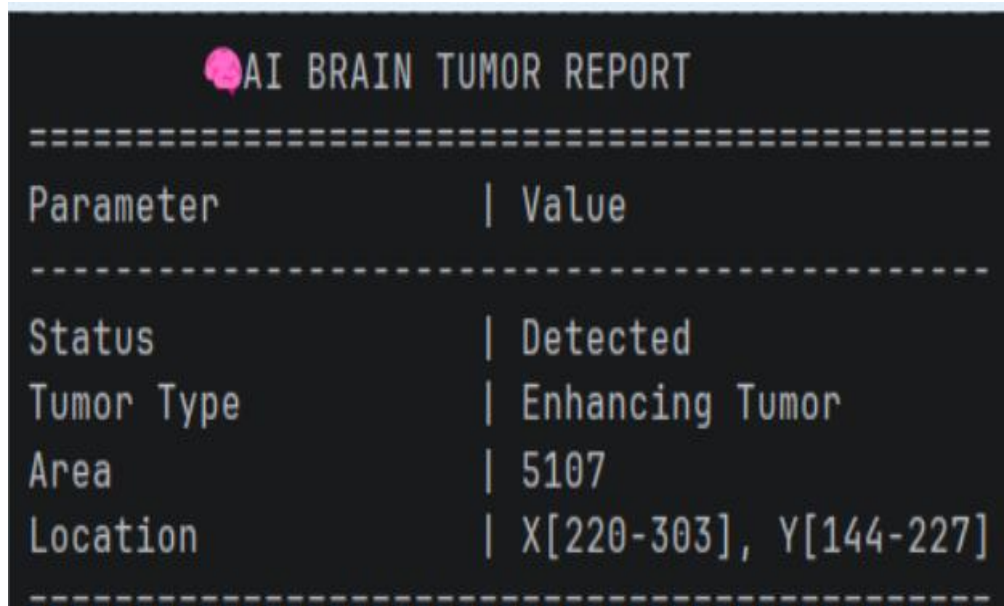


Fig. 5.6 Tumor Detection Output

5.5.7 3D Tumor Detection Result

The 3D Tumor Detection Result section presents a volumetric representation of the detected tumor regions to provide enhanced visualization and spatial understanding. The segmented tumor is reconstructed across multiple MRI slices to form a three-dimensional view, allowing better interpretation of tumor size, shape, and location within the brain. This representation helps in identifying the extent of tumor spread more effectively compared to 2D images.

The 3D visualization improves clinical analysis by enabling doctors to observe tumor boundaries from different perspectives. It also supports better treatment planning by providing a comprehensive view of tumor structures such as Edema, Tumor Core, and Enhancing Tumor. Fig. 5.7 illustrates the 3D Tumor Detection Result, demonstrating the enhanced visualization capability of the system.



The image shows a terminal-style report titled "AI BRAIN TUMOR REPORT" with a pink brain icon. The report contains a table with four rows of tumor detection data. The table is framed by dashed lines. The data indicates a detected enhancing tumor with an area of 5107 and a specific location in X and Y coordinates.

Parameter	Value
Status	Detected
Tumor Type	Enhancing Tumor
Area	5107
Location	X[220-303], Y[144-227]

Fig. 5.7 3D Detection Report

5.5.8 Doctor Login

The Doctor Login provides healthcare professionals with secure access to patient records, MRI scans, and tumor detection results. It enables doctors to analyze diagnostic reports, provide medical recommendations, and manage appointments efficiently within the system. Fig. 5.8 illustrates the Doctor Login interface, which is designed to support smooth navigation and quick access to essential functionalities.

Furthermore, the interface incorporates authentication mechanisms to ensure that only authorized users can access sensitive medical data. It facilitates effective communication between doctors and patients by allowing timely review of results and feedback. This enhances clinical decision-making and contributes to improved healthcare delivery and patient management.

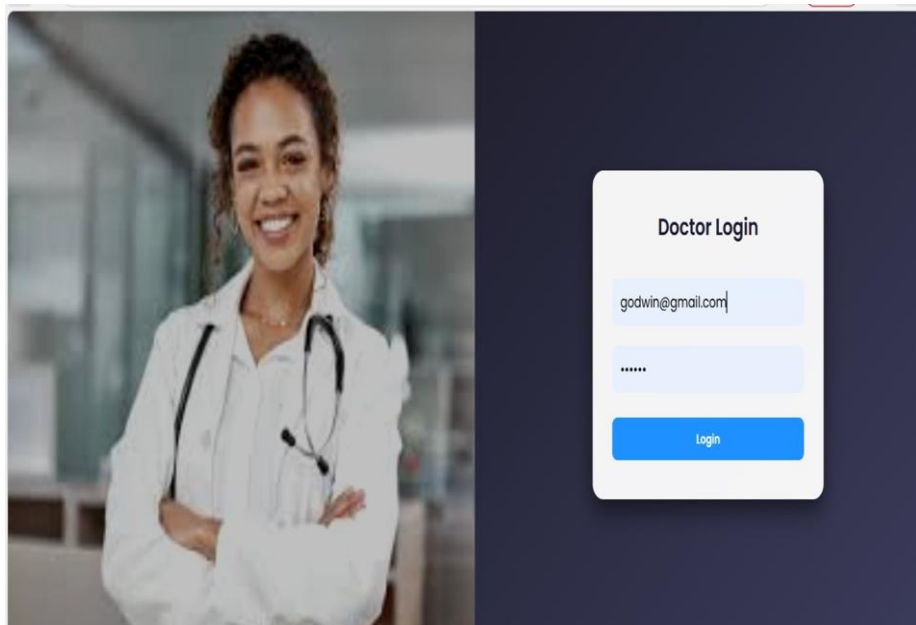


Fig. 5.8 Doctor Login

5.5.9 Admin Dashboard

The Admin Dashboard enables administrators to manage and monitor the overall functioning of the system. It provides comprehensive control over users, data, and system configurations. Fig. 5.9 illustrates the Admin Dashboard. This module ensures efficient system administration and maintains data integrity and security.

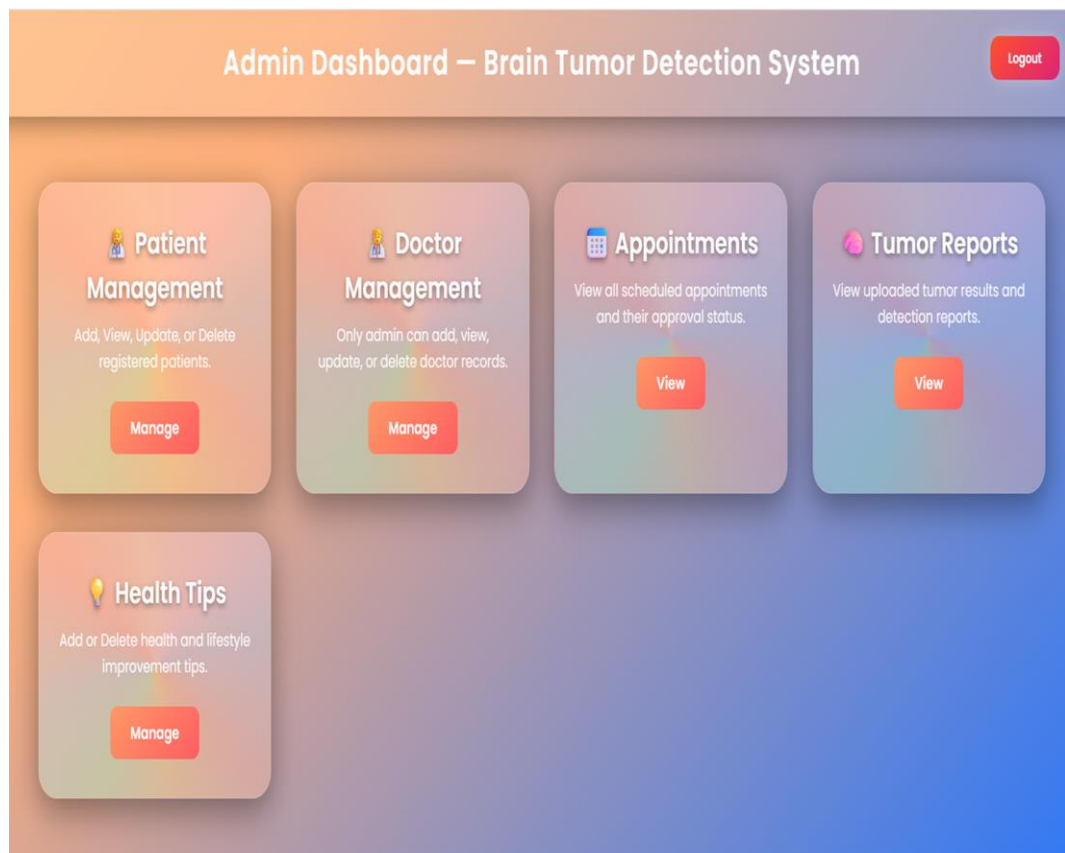


Fig. 5.9 Admin Dashboard

5.5.10 Appointment Management Interface

The Appointment Management Interface facilitates communication between patients and doctors by enabling online appointment scheduling. It streamlines the consultation process and improves healthcare coordination. Fig. 5.10 shows the Appointment Management Interface. This module maintains accurate records of appointments and enhances the efficiency of medical services.

Book Your Appointment

06 - 04 - 2026

18 : 47

Fever

Book Appointment

← Back to Dashboard

Go to Tumor Detection

All Booked Appointments

Appointment ID	Patient Name	Patient Email	Patient Mobile	Doctor Name	Doctor Email	Doctor Mobile	Specialization	Date	Time	Reason	Status	Actions
4	Angelin Shiny.L	shinyy@gmail.com	8667442783	N/A	N/A	N/A	N/A	2026-04-06	18:47:00	Fever	Pending	Update Delete
2	priya	priya@gmail.com	8476576586	N/A	N/A	N/A	N/A	2026-03-06	12:22:00	Vomit,fever	Pending	Update Delete
3	Ranishiny	shiny@gmail.com	9873472324	N/A	N/A	N/A	N/A	2026-02-20	17:47:00	High Fever	Pending	Update Delete

Fig. 5.10 Appointment Management Page

5.5.11 Tumor Detection Report Page

The Tumor Detection Report Page generates detailed diagnostic reports based on AI predictions. These reports include segmentation results and relevant patient information for medical consultation. Fig. 5.10 illustrates the Tumor Detection Report Page. The interface ensures clarity and professionalism, enabling users to download and print reports for future reference.

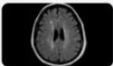
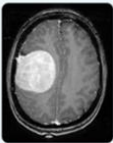
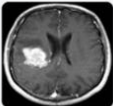
				
Patient Name	Result	Image	Created At	Actions
shiny	No Tumor Detected		2025-10-26 17:31:22	Delete
priya	Tumor Detected		2025-10-21 23:13:07	Delete
Angelin	Tumor Detected		2025-10-21 22:42:12	Delete

Fig. 5.11 Tumor Detection Report

5.5.12 Health Tips Page

The Health Tips Page is an informative module integrated into the Brain Tumor Detection and Segmentation System to promote awareness and preventive healthcare. It provides essential information related to brain health, early detection of brain tumors, and general wellness practices. This page aims to educate users and encourage timely medical consultation, thereby enhancing the overall effectiveness of the healthcare portal.

The interface is designed with clarity and accessibility, ensuring that users can easily understand and benefit from the information provided. Fig. 5.11 illustrates the Health Tips Page of the Brain Tumor Detection and Segmentation System. The content is managed by the administrator and stored securely in the MySQL database, allowing regular updates and ensuring the reliability of medical guidance. This module complements the AI-based detection system by fostering patient awareness and supporting proactive healthcare.

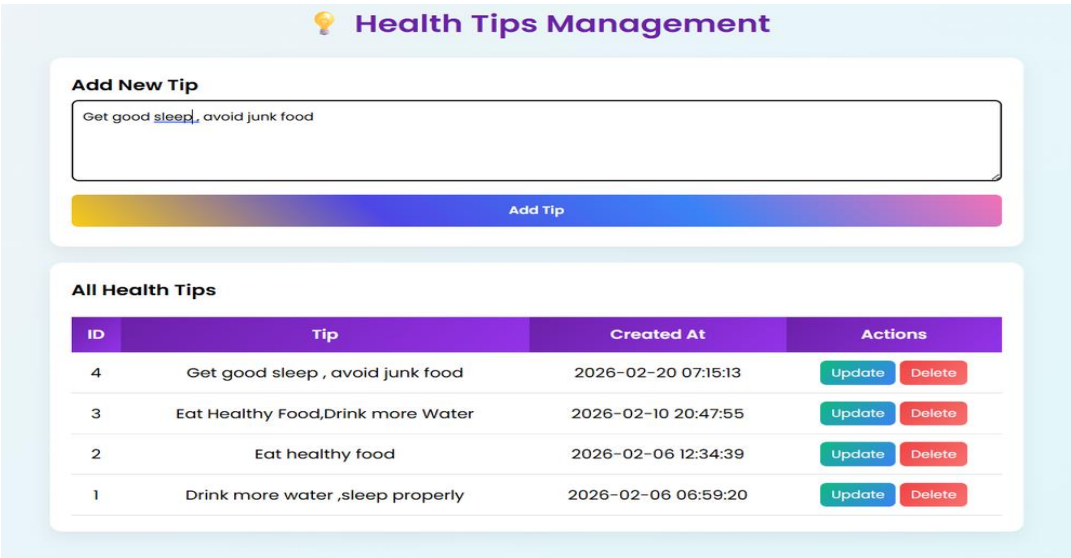


Fig. 5.12 Health Tips Page

6 METHODOLOGY

6.1 Methodology Overview

This chapter presents the methodology and implementation of the proposed Deep Learning-Based Brain Tumor Detection and Segmentation System with an Integrated Doctor–Patient Portal. It provides a comprehensive description of the systematic procedures followed to design, develop, and deploy the system. The methodology integrates advanced artificial intelligence techniques with modern web technologies to automate brain tumor detection from MRI scans, ensuring accuracy, efficiency, and reliability in medical diagnosis. The proposed approach reduces manual effort, enhances clinical decision-making, and improves accessibility to healthcare services through a centralized web portal.

CNN, U-Net, and Attention U-Net were implemented and compared, and only the Attention U-Net model was selected and integrated with the web-based portal based on its superior performance. These models are trained using the Brain Tumor Segmentation (BraTS) dataset, which is widely recognized in the medical imaging domain. The trained models are deployed and integrated into a web-based environment using a Flask API, enabling real-time tumor detection and report generation. This chapter elaborates on each phase of development, including data acquisition, preprocessing, model training, evaluation, system integration, and implementation.

The methodology of the proposed system follows a structured and sequential approach to ensure precision, efficiency, and reliability. It begins with data acquisition from the BraTS dataset, followed by preprocessing, model development, training, and evaluation. The trained models are then integrated with a web-based portal through a Flask API to provide automated diagnostic services. Each phase contributes significantly to achieving accurate tumor detection and seamless interaction among patients, doctors, and administrators. This systematic approach ensures the smooth flow of data from MRI acquisition to result generation and presentation through the system and downloadable PDF reports.

6.2 Data Acquisition

Data acquisition is the foundational stage of the proposed methodology. The system utilizes the Brain Tumor Segmentation (BraTS) dataset, which contains high-quality multimodal MRI images along with expert-annotated ground truth segmentation masks. This dataset is widely used in medical image analysis and provides a reliable benchmark for training deep learning models.

The BraTS dataset includes four MRI modalities—T1-weighted, T1-contrast enhanced (T1ce), T2-weighted, and Fluid Attenuated Inversion Recovery (FLAIR). Each modality provides unique information about brain tissues and tumor characteristics. The dataset also categorizes tumor regions into Edema (ED), Tumor Core (TC), and Enhancing Tumor (ET), enabling accurate and supervised learning.

The BraTS dataset is widely utilized in the proposed system due to its comprehensive and high-quality medical imaging data. It provides multimodal MRI scans that enable detailed and accurate tumor analysis by combining different imaging modalities. Expert-annotated ground truth segmentation masks are included, which are used to train and validate deep learning models for precise tumor detection. High-resolution three-dimensional medical images are provided, allowing detailed visualization of tumor structures. The dataset is recognized as a standard benchmark in brain tumor research, ensuring reliability and consistency in model evaluation. It is provided in NIfTI (.nii) format, which is suitable for medical imaging applications and facilitates efficient data handling.

6.3 Data Preprocessing

Data preprocessing is an essential step that enhances the quality, consistency, and usability of MRI images before they are used for model training. Medical

images often contain noise, intensity variations, and inconsistencies that can negatively impact model performance. Therefore, preprocessing ensures uniformity and improves segmentation accuracy.

The preprocessing phase involves resizing images, normalizing pixel values, reducing noise, and preparing segmentation masks. Additionally, data augmentation techniques are applied to improve model generalization and prevent overfitting. These steps significantly enhance the performance and reliability of the deep learning models.

6.3.1 Preprocessing Techniques

Various preprocessing techniques are applied to MRI images to improve the quality and consistency of data before it is provided to the deep learning models. Image resizing is performed to standardize all images into fixed dimensions such as 128×128 or 256×256 pixels, ensuring uniform input size for model training. Normalization is applied to scale pixel intensity values between 0 and 1, thereby improving numerical stability during model processing. Z-score standardization is also performed to normalize intensity values based on their mean and standard deviation, which helps in reducing intensity variations across different scans.

Noise reduction techniques are applied to remove unwanted distortions and artifacts present in MRI images, thereby enhancing image quality. Data augmentation methods such as rotation, flipping, zooming, and translation are applied to increase dataset diversity and improve model generalization. Slice extraction is performed to convert three-dimensional MRI volumes into two-dimensional image slices, which are suitable for model training. Mask preparation is carried out to generate binary and multiclass segmentation labels corresponding to tumor regions. Intensity standardization is applied to ensure uniform contrast across images; while cropping and padding techniques are used to focus on the region of interest and maintain consistent image dimensions.

6.3.2 Benefits of Preprocessing

The application of preprocessing techniques significantly enhances the performance of the deep learning models used in the system. Model accuracy and convergence are improved by providing clean and standardized input data. Image clarity and consistency are enhanced, enabling better feature extraction by the model. Overfitting and training errors are reduced through the use of augmentation and normalization techniques. Furthermore, standardized input data is ensured for all images, which improves the reliability and effectiveness of tumor detection and segmentation results.

6.4 Model Development

The model development phase involves designing and implementing deep learning architectures capable of detecting and segmenting brain tumors with high precision. The proposed system utilizes CNN, U-Net, and Attention U-Net models due to their effectiveness in medical image analysis and segmentation tasks.

6.4.1 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are utilized for feature extraction from MRI images in the proposed system. Through this approach, spatial hierarchies and complex patterns are automatically learned, making CNNs highly effective for tumor detection. Significant features are extracted from MRI images to capture important structural information. Complex patterns and textures within the images are identified to support accurate analysis. Segmentation accuracy is enhanced through robust feature representation, and support is provided to advanced segmentation models such as U-Net and Attention U-Net.

6.4.2 U-Net Architecture

U-Net is implemented as a specialized convolutional neural network designed for biomedical image segmentation. It consists of an encoder–decoder architecture with skip connections that preserve spatial information during processing. Precise pixel-level segmentation is achieved by combining high-level and low-level features. Effective performance is maintained even with limited datasets, making it suitable for medical imaging applications. Fine details within MRI images are preserved, and high-resolution segmentation outputs are generated to support accurate tumor identification.

6.4.3 Attention U-Net

Attention U-Net is utilized as an enhanced version of the U-Net architecture by incorporating attention mechanisms. These mechanisms focus on relevant tumor regions while suppressing irrelevant background information. Tumor localization is improved by highlighting important regions within MRI scans. False positives are reduced by minimizing unnecessary feature activation. Segmentation accuracy is further enhanced, and overall diagnostic reliability is increased through improved model focus and precision.

6.4.4 Comparative Analysis of U-Net and Attention U-Net

A comparative analysis is performed between the U-Net and Attention U-Net architectures to evaluate their effectiveness in brain tumor segmentation. Both models are widely used in medical image processing due to their ability to perform precise pixel-level segmentation. U-Net is designed with an encoder–decoder structure and skip connections, which help in preserving spatial information and producing high-resolution segmentation outputs. It performs efficiently even with limited datasets and is capable of capturing fine details in MRI images.

Attention U-Net extends the standard U-Net architecture by incorporating attention mechanisms that enable the model to focus on relevant regions of interest,

particularly tumor areas, while suppressing background noise. This selective focus improves the model's ability to localize tumors more accurately. As a result, Attention U-Net enhances segmentation performance by reducing false positives and improving boundary detection of tumor regions.

In comparison, U-Net provides strong baseline performance with faster computation and lower complexity, making it suitable for general segmentation tasks. However, Attention U-Net demonstrates superior performance in complex medical imaging scenarios by improving feature selection and enhancing tumor localization. Although it requires slightly higher computational resources, it produces more accurate and reliable segmentation results.

Overall, it is observed that both models are effective for brain tumor detection; however, Attention U-Net achieves better accuracy and precision due to its ability to focus on critical features within MRI images. Therefore, Attention U-Net is considered more suitable for advanced tumor segmentation tasks in the proposed system. Based on this comparative analysis, Attention U-Net is selected as the final model due to its higher Dice Score and improved tumor localization capability.

6.5 Tumor Segmentation and Classification

Tumor segmentation is performed to identify and isolate abnormal tissues within MRI scans. In the proposed system, tumor regions are classified into Edema, Tumor Core, and Enhancing Tumor classification supports healthcare professionals in accurate diagnosis and treatment planning. Precise localization of tumor regions is achieved through segmentation techniques. Color-coded visualization of segmentation results is provided for better interpretation. Clinical decision-making is improved, and manual analysis is reduced.

6.6 Model Training and Validation

The deep learning models are trained using the preprocessed BraTS dataset to ensure accurate tumor detection and segmentation. The dataset is divided into training, validation, and testing subsets to enable unbiased evaluation and optimal model performance. The Adam optimizer is utilized with a learning rate of 0.0001 to achieve stable convergence. Training is conducted over 50 to 100 epochs with a batch size of 8 to 16. Loss functions such as Dice Loss and Binary Cross-Entropy are applied to improve segmentation accuracy. Model performance is evaluated using metrics including **Dice Score**, Accuracy, Precision, Recall, F1-Score, and Intersection over Union (IoU) to ensure reliability and effectiveness.

6.7 Integration Using Flask API

The trained deep learning models are integrated with the web portal using the Flask framework. Flask acts as an intermediary between the Python-based AI models and the PHP-based web application. MRI images are received from the web portal through API requests. These images are processed using the trained models to generate prediction results. The results are returned in real time, displayed through the system interface, and included in downloadable PDF reports. Communication between the artificial intelligence module and the web application is thereby enabled efficiently.

6.8 Web Portal Implementation

The web portal is developed to provide an interactive interface for patients, doctors, and administrators. It is implemented using HTML, CSS, JavaScript, and PHP to ensure responsive and dynamic functionality. Secure user authentication is implemented to control access to system features. MRI image upload and tumor detection report generation functionalities are provided for diagnostic purposes.

6.9 Database Implementation

The system utilizes MySQL as the database management system to store and manage all healthcare-related data securely. Patient records, MRI scans, tumor detection results, and appointment details are maintained within structured database tables. Key tables such as Patients, Doctors, Admin, MRI_Scans, Tumor_Reports, Appointments, and Health_Tips are created and properly linked to ensure data consistency and integrity. Efficient data retrieval, storage, and management are achieved through this structured database design.

6.10 System Implementation Environment

The proposed system is developed using modern tools and technologies to ensure efficiency and reliability. Python is used for implementing the artificial intelligence models, with TensorFlow and Keras serving as the core frameworks for brain tumor detection and segmentation. The web application is developed using PHP for the backend and HTML, CSS, and JavaScript for the frontend, providing an interactive user interface. Integration between the AI module and the web portal is achieved through a Flask API, enabling seamless communication and real-time processing.

MySQL is used for secure storage and management of patient and diagnostic data, while WAMP Server provides the local development environment. The deep learning models are trained and evaluated using the BraTS dataset to ensure reliable results. Development and testing are carried out using tools such as PyCharm to support efficient implementation.

7 SYSTEM TESTING

7.1 Testing Overview

System Testing and Validation constitute a vital phase in the Software Development Life Cycle (SDLC), ensuring that the developed application meets its intended objectives and performs reliably in real-world scenarios. This chapter presents a comprehensive evaluation of the proposed Deep Learning-Based Brain Tumor Detection and Segmentation System with an Integrated Doctor–Patient–Admin Portal. The primary aim of this phase is to verify that all system components function accurately, efficiently, and in accordance with the specified functional and non-functional requirements.

The proposed system integrates advanced deep learning techniques with modern web technologies to provide automated and precise detection of brain tumors from MRI scans. It employs Convolutional Neural Networks (CNN), U-Net, and Attention U-Net models trained on the BraTS dataset for accurate tumor segmentation. In addition, the system incorporates a web-based platform developed using HTML, CSS, JavaScript, PHP, and MySQL, with Flask serving as the bridge between the artificial intelligence module and the user interface. Therefore, rigorous testing was essential to validate the seamless interaction among these components.

Comprehensive testing was conducted to ensure the correctness, reliability, security, and performance of the system. Various testing techniques, including unit testing, integration testing, functional testing, performance testing, security testing, database testing, and user acceptance testing, were employed. These tests confirmed the system's ability to process MRI images accurately, generate reliable tumor detection results, manage patient and medical data securely, and provide an intuitive user experience for patients, doctors, and administrators.

The results obtained from the testing process demonstrate that the proposed system operates efficiently and meets the desired quality standards. The artificial intelligence module produced accurate segmentation outputs, while the integrated web portal ensured real-time prediction, secure data handling, and effective communication among stakeholders. This chapter provides a detailed discussion of

the testing methodologies, tools, test cases, and validation outcomes, thereby establishing the reliability and practical applicability of the system in modern healthcare environments.

7.2 Objectives of System Testing

The primary objective of system testing is to ensure that the developed Brain Tumor Detection and Segmentation System operate accurately, efficiently, and reliably in accordance with the specified requirements. Testing verifies that all system components function correctly both individually and collectively, thereby guaranteeing the quality and robustness of the final product. It also helps identify potential defects and ensures that the system delivers precise diagnostic results.

In this project, system testing plays a vital role in validating the integration of artificial intelligence with web technologies. Since the system processes sensitive medical data and generates clinical insights, it is essential to ensure high accuracy, security, and performance. The testing phase confirms that the system can detect brain tumors from MRI scans, generate segmentation outputs, and store patient records securely while maintaining optimal performance.

Furthermore, system testing ensures seamless communication between various modules, including the AI detection module, web portal, database, and Flask API. It evaluates the system's usability, responsiveness, and scalability, ensuring that it meets the expectations of patients, doctors, and administrators. The comprehensive testing approach guarantees that the system is reliable and suitable for real-world healthcare applications.

7.3 Testing Strategy and Methodology

A systematic and well-structured testing strategy was adopted to evaluate the overall performance and reliability of the proposed system. The testing process

followed a step-by-step approach, beginning with unit testing of individual modules and progressing to integration, functional, and system testing. This ensured that each component of the system operated efficiently and contributed to the seamless functioning of the complete application.

Both manual and automated testing techniques were employed to validate different aspects of the system. Manual testing was used to verify the functionality and usability of the web portal, while automated testing tools were utilized to evaluate the performance of the deep learning models. This hybrid approach ensured comprehensive validation of the system and improved its reliability and robustness.

The testing methodology also focused on evaluating the communication between the Flask API and the PHP-based web portal. Special attention was given to validating the transmission of MRI images, processing of data by the AI models, and generation of tumor detection reports. This approach ensured accurate real-time predictions and smooth data flow across all system components.

7.4 Unit Testing

Unit testing was conducted to verify the functionality of individual modules within the proposed system. Each component was tested independently to ensure it performed its intended tasks accurately and efficiently. This phase helped identify and rectify errors at an early stage, thereby improving the overall quality and reliability of the system.

The Patient Module was tested to verify the correctness of patient registration and login validation processes. The functionality for uploading MRI scans was evaluated to ensure successful image submission and transmission to the detection module. The ability to view tumor detection reports was also tested to confirm accurate display of results on the patient dashboard.

The Doctor Module was tested to ensure secure login authentication and proper access to the doctor dashboard. The functionality for viewing patient reports and segmentation results was validated to confirm accurate data retrieval. Appointment management and consultation features were tested to ensure smooth interaction between doctors and patients.

The Admin Module was evaluated to verify administrative control over system operations. Functionalities such as managing patient and doctor records were tested to ensure accurate data handling. Monitoring of tumor detection reports and database records was validated, and overall system control features were tested to confirm effective administration.

The Detection (AI) Module was tested to evaluate preprocessing techniques, including image resizing, normalization, and noise reduction. Tumor detection and segmentation were validated using Attention U-Net models to ensure accurate prediction outputs. The generation of segmentation masks and classification results was verified for correctness and consistency.

The tools used for testing included PyCharm for AI module validation, WAMP Server for web application hosting, MySQL for database management, and web browsers for interface testing. The results of the testing process indicated that all modules performed accurately and met the expected outcomes. The system demonstrated reliable performance, correct functionality, and successful integration of all components.

7.5 Integration Testing

Integration testing was conducted to verify the seamless communication and coordination between various components of the proposed system. This phase ensured that all modules, including the web portal, artificial intelligence models, Flask API, and database, operated together without errors while maintaining data

consistency and system reliability. The objective of this testing was to validate the interaction between integrated components and to ensure that data flow across modules was accurate and efficient.

The integration between the PHP-based web portal and the MySQL database was tested to ensure proper data storage, retrieval, and management. Communication between the web portal and the Flask API was validated to confirm that MRI images were successfully transmitted for processing. The interaction between the Flask API and the trained AI model (.h5 file) was tested to ensure that prediction results were generated accurately. The process of MRI image upload and its transfer to the tumor detection module was verified to ensure correct data flow. The results of integration testing indicated that all modules were successfully integrated and functioned cohesively. Accurate and real-time tumor detection was achieved through effective communication between system components. The system demonstrated reliable performance, efficient data exchange, and consistency across all integrated modules.

7.6 Functional Testing

Table 7.1 presents the results of functional testing conducted to verify that each feature of the system operates in accordance with the specified requirements. During this phase, the outputs generated by the system were systematically compared with the expected results to ensure accuracy, correctness, and reliability of all functionalities.

Table 7.1 Functional Test Cases

Test Case ID	Description	Expected Output	Status
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TC01	Patient Registration	Successful registration	Pass
TC02	Patient Login	Login authenticated	Pass
TC03	MRI Image Upload	Image uploaded successfully	Pass
TC04	Tumor Detection	Segmentation result generated	Pass
TC05	Report Generation	Tumor report displayed	Pass
TC06	Appointment Booking	Appointment scheduled	Pass
TC07	Doctor Login	Access granted	Pass
TC08	Admin Login	Dashboard displayed	Pass
TC09	Flask API Integration	Prediction returned successfully	Pass

7.7 AI Model Testing and Validation

The performance of the deep learning models was evaluated using the BraTS dataset to assess their effectiveness in brain tumor detection and segmentation. The trained models demonstrated high accuracy in identifying tumor regions and producing reliable segmentation outputs. Among the implemented models, Attention U-Net exhibited superior performance due to its ability to focus on relevant tumor regions while minimizing the influence of background noise.

The evaluation of model performance was carried out using standard metrics, including Accuracy, Dice Score, Precision, Recall, Intersection over Union (IoU), and F1-Score. These metrics were used to measure the correctness, overlap, and reliability of the predicted segmentation results in comparison with the ground truth data. The results obtained from these metrics confirmed the effectiveness of the models in handling complex medical imaging data. The observations from the evaluation are presented in Table 7.2, where it is shown that Attention U-Net achieved the highest accuracy and superior segmentation performance compared to other models.

Table 7.2 AI Model Performance

Model	Accuracy	Dice Score	Precision	Recall
CNN	94.2%	0.89	0.91	0.88
U-Net	96.8%	0.92	0.94	0.91
Attention U-Net	98.5%	0.95	0.97	0.94

7.8 Database Testing

Database testing was conducted to ensure the accuracy, consistency, and reliability of the data stored within the system. The MySQL database was validated to confirm the correctness of table structures, relationships, and integrity constraints. Verification of table schemas and relationships was performed to ensure proper organization of data. Primary and foreign keys were validated to maintain referential integrity and prevent inconsistencies. CRUD (Create, Read, Update, Delete) operations were tested to ensure efficient data manipulation within the system.

The storage and retrieval of MRI scans and tumor detection reports were verified to confirm accurate data handling. Mechanisms were also tested to prevent duplicate and inconsistent data entries within the database. The results of database testing indicated that the system operated efficiently and

maintained data integrity. All database operations were executed successfully, ensuring reliable storage and retrieval of information across system modules.

7.9 User Interface Testing

User Interface (UI) testing was conducted to evaluate the usability, accessibility, and responsiveness of the web portal. The system interface was designed to provide a smooth and intuitive user experience for patients, doctors, and administrators. Navigation and layout structures were verified to ensure ease of use and logical flow across different modules. Input forms and interactive elements such as buttons were validated to confirm proper functionality and error handling. The responsiveness of the interface was assessed to ensure compatibility across different screen sizes and devices. Cross-browser compatibility testing was performed to verify consistent performance across various web browsers.

Additionally, overall user-friendliness was evaluated to ensure that the system could be easily used by individuals with varying levels of technical expertise. The results of UI testing indicated that the interface was user-friendly, responsive, and functionally reliable. All interface components operated as expected, providing an effective and accessible user experience.

7.10 Performance Testing

Table 7.3 presents the results of performance testing conducted to evaluate the responsiveness, efficiency, and stability of the system under real-time conditions. During this phase, the system performance was assessed

while processing MRI images and generating diagnostic reports. The results indicated that the system demonstrated reliable and stable performance, with efficient execution of MRI analysis and timely report generation.

Table 7.3 Performance Metrics

Parameter	Result
Average Prediction Time	3–5 seconds
System Response Time	Less than 5 seconds
Database Retrieval Speed	Fast
System Stability	High
Overall Performance	Efficient

7.11 Security Testing

Security testing was conducted to ensure the protection of sensitive medical data and to prevent unauthorized access to the system. Appropriate authentication and authorization mechanisms were implemented to safeguard system integrity and maintain confidentiality. Secure login authentication was

enforced to validate user identity before granting access. Role-based access control was implemented to restrict system functionalities based on user roles such as patient, doctor, and administrator.

Password encryption techniques were applied to protect user credentials from potential security threats. Input validation mechanisms were incorporated to prevent SQL injection and other malicious attacks. Additionally, secure storage practices were followed to protect patient records and sensitive healthcare information within the database. The results of security testing indicated that the system ensured secure handling of both medical and user data.

7.12 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted to evaluate the system from the perspective of end users, including patients, doctors, and administrators. Real-time scenarios were simulated to assess system usability, functionality, and overall effectiveness in a practical environment. The system was tested to ensure that all features operated as expected and met user requirements.

The outcomes of UAT demonstrated that accurate tumor detection and segmentation were achieved through the integrated AI module. Efficient appointment scheduling and report management functionalities were validated to support smooth healthcare operations. Seamless integration between the artificial intelligence module and the web portal was confirmed. The interface was found to be easy to navigate and user-friendly, enabling users to interact with the system effectively.

7.13 Summary of Tests Conducted

Table 7.4 presents the summary of tests conducted. The proposed system was rigorously tested using unit testing, integration testing, system testing, and user acceptance testing to ensure reliability and accuracy. Each module, including the AI-based tumor detection, web portal, and database, functioned efficiently and met the specified requirements. The deep learning models demonstrated accurate segmentation and classification of brain tumors using MRI images from the BraTS dataset. The test results confirmed that the system operates securely, efficiently, and effectively, satisfying both functional and performance criteria.

Table 7.4 Summary of Tests Conducted

Test Type	Purpose	Status
Unit Testing	Verified individual modules	Passed
Integration Testing	Ensured module connectivity	Passed
Functional Testing	Validated system features	Passed
AI Model Testing	Evaluated model accuracy	Passed

Table 7.4 Summary of Tests Conducted (Continues)

Database Testing	Verified data integrity	Passed
User Interface Testing	Assessed usability	Passed
Performance Testing	Evaluated efficiency	Passed
Security Testing	Ensured data protection	Passed
User Acceptance Testing	Confirmed user satisfaction	Passed

8 CONCLUSION

The primary objective of this project was to develop an automated and intelligent system capable of detecting and segmenting brain tumors from MRI scans using deep learning techniques. This objective has been successfully achieved through the implementation of Convolutional Neural Networks (CNN), U-Net, and Attention U-Net architectures. These models were comparatively evaluated to identify the most effective approach for tumor segmentation, and based on performance metrics, Attention U-Net was selected as the final model. These models were trained using the Brain Tumor Segmentation (BraTS) dataset, ensuring high accuracy and reliability in tumor identification and segmentation.

The proposed system effectively preprocesses MRI images, extracts meaningful features, and generates precise segmentation outputs. Among the implemented models, the Attention U-Net demonstrated superior performance due to its ability to focus on relevant tumor regions, thereby improving segmentation accuracy and reducing false positives. The model achieved improved Dice Score values, indicating better overlap between predicted and ground truth tumor regions. The evaluation metrics, including Dice Score, Accuracy, Precision, and Recall, validated the effectiveness of the developed models.

In addition to the artificial intelligence component, a user-friendly web portal was developed using HTML, CSS, JavaScript, PHP, and MySQL. The Flask framework was employed to integrate the trained deep learning models with the web application, enabling real-time tumor detection and report

generation. The final segmentation results generated using the Attention U-Net model are displayed through the portal and provided as downloadable PDF reports for clinical reference. The portal facilitates seamless interaction among patients, doctors, and administrators, allowing secure data management, appointment scheduling, and efficient communication.

The database was designed using MySQL to store patient records, MRI scans, tumor detection results, and appointment details securely. Rigorous system testing and validation confirmed the reliability, accuracy, and performance of the system. The integration of artificial intelligence with web technologies enhances diagnostic efficiency, reduces manual workload, and supports medical professionals.

Overall, the proposed Brain Tumor Detection and Segmentation System demonstrate significant potential in assisting healthcare professionals by providing accurate, efficient, and automated diagnostic solutions. It represents a valuable contribution to the domains of medical imaging, artificial intelligence, and healthcare informatics.

8.1 Key Achievements of the Project

The proposed system has successfully achieved several significant milestones in the domain of intelligent healthcare applications. An automated brain tumor detection and segmentation system has been developed using advanced deep learning techniques. Convolutional Neural Networks (CNN), U-Net, and Attention U-Net architectures have been implemented to perform accurate tumor analysis. The BraTS dataset has been utilized for effective model training and evaluation, resulting in high accuracy in tumor detection

and segmentation. The trained AI models have been successfully integrated with a Flask-based web application, enabling real-time prediction and analysis.

A comprehensive **Doctor–Patient portal** has been designed and developed to support efficient healthcare management. A secure MySQL database has been implemented to store and manage medical data reliably. Real-time MRI analysis and report generation functionalities have been enabled within the system. Data security has been ensured through authentication mechanisms and role-based access control. Furthermore, the system has been thoroughly validated through comprehensive testing, confirming its accuracy, reliability, and overall performance.

8.2 Applications of the Proposed System

The proposed system has wide-ranging applications in both medical and research domains due to its ability to provide automated and accurate tumor detection. It can be utilized in hospitals and diagnostic centers for early detection of brain tumors, thereby supporting timely medical intervention. Radiology departments can benefit from automated MRI analysis, reducing manual workload and improving efficiency. The system can also be applied in telemedicine platforms to enable remote diagnosis and consultation.

In addition, the system can be used in medical research institutions for neuroimaging studies and advanced research in brain tumor analysis. Educational institutions can utilize the system for teaching and research in artificial intelligence and medical imaging. Healthcare organizations can adopt the system to improve diagnostic efficiency and patient care. It can also

be integrated into AI-driven clinical decision support systems and utilized for medical data analysis and predictive healthcare applications.

8.3 Limitations of the Proposed System

Despite its effectiveness, the proposed system has certain limitations that can be addressed in future research and development. High computational resources are required for training deep learning models, which may limit accessibility in resource-constrained environments. The system is dependent on the quality and availability of MRI datasets, which can affect model performance.

It is currently limited to MRI-based brain tumor detection. The performance of the system may vary when tested with unseen or low-quality images. Clinical validation is required before real-world deployment to ensure medical reliability and acceptance. The system primarily operates in a controlled environment, which may limit its adaptability in real-time clinical settings.

8.4 Future Scope

The proposed system offers significant opportunities for enhancement with advancements in artificial intelligence and healthcare technologies. Future improvements can include deployment on cloud platforms such as AWS, Microsoft Azure, or Google Cloud to enhance scalability and accessibility. A mobile application can be developed to provide remote access to the system. Advanced techniques such as 3D brain tumor segmentation can be implemented to improve diagnostic precision.

Integration with Hospital Information Systems (HIS) and Electronic Health Records (EHR) can be achieved to enable seamless data exchange. Explainable Artificial Intelligence (XAI) can be incorporated to improve transparency and trust in model predictions. The system can be expanded to detect other neurological disorders such as Alzheimer's and Parkinson's disease. The use of larger and more diverse datasets can further enhance model accuracy.

Additional features such as real-time telemedicine and remote consultation can be implemented to improve healthcare accessibility. Automated generation of downloadable medical reports in PDF format can be incorporated for user convenience. Blockchain technology can be integrated to ensure secure medical data sharing. Support for multilingual interfaces can be provided to enhance usability for diverse users. IoT-enabled medical imaging solutions can be developed to enable smart healthcare systems. Optimization using advanced deep learning models such as ResNet, DenseNet, and Vision Transformers can further improve performance. Finally, the system can be deployed as a Software-as-a-Service (SaaS) healthcare platform to provide scalable and accessible medical solutions.

APPENDIX

CODE

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Brain Tumor Detection Portal</title>
<style>
:root {
  --primary: #1e3d59;
  --secondary: #f5a623;
  --accent: #caf0f8;
  --dark: #03045e;
  --light: #f8f9fa;
  --radius: 16px;
  --shadow: 0 10px 30px rgba(0,0,0,0.1);
  font-family: 'Poppins', sans-serif;
}
* { margin:0; padding:0; box-sizing:border-box; scroll-behavior:smooth; }
body { color: var(--dark); overflow-x: hidden; background: var(--light); }
/* Header */
header {
```



```

position: fixed; top:0; left:0; right:0;
display:flex; justify-content:space-between; align-items:center;
padding:1rem 3rem; background: rgba(30,61,89,0.85); z-index:100;
border-bottom: 1px solid rgba(255,255,255,0.2);
}
header:hover { background: rgba(30,61,89,1); }
.logo { color:#fff; font-size:1.6rem; font-weight:700; text-decoration:none;
text-shadow:0 2px 8px rgba(0,0,0,0.4); }
.logo span { color: var(--secondary); }
/* Top Buttons */
.nav-btns { display:flex; gap:1rem; }
.nav-btns button {
background: var(--secondary); border:none; color:#fff;
padding:10px 18px; border-radius:8px; font-weight:600; cursor:pointer;
transition:0.3s;
}
.nav-btns button:hover { background: var(--primary); }
/* Hero Section */
.hero {
height:100vh; display:flex; align-items:center; justify-content:center; text-
align:center;
color:white; background: linear-gradient(rgba(30,61,89,0.7),
rgba(30,61,89,0.7)),
url('images/brain.jpg') center/cover no-repeat;
padding: 6rem 2rem;
}
.hero-content { max-width:800px; animation: fadeIn 1.5s ease; }
.hero-content h1 { font-size:3rem; line-height:1.3; margin-bottom:1rem;

```

```

text-shadow:0 3px 10px rgba(0,0,0,0.4); }
.hero-content h1 span { color: var(--secondary); }
.hero-content p { font-size:1.1rem; color:#eaf6ff; margin-bottom:2rem; }
.hero-content .btn { display:inline-block; background: var(--secondary);
color:white; padding:0.9rem 1.8rem; margin-right:0.8rem; border-radius:
16px; font-weight:600; text-decoration:none; transition: all 0.3s; box-
shadow: 0 6px 14px rgba(0,0,0,0.2); }
.hero-content .btn:hover { background: var(--primary); transform:
translateY(-3px); }
/* Features */
section.features { padding:5rem 1.5rem; text-align:center;
background:#ffffff; }
.features h2 { font-size:2rem; color: var(--dark); margin-bottom:2rem;
position:relative; }
.features h2::after { content:""; display:block; height:3px; width:80px;
background: var(--primary); margin:1rem auto 0; border-radius:3px; }
.features .grid { display:grid; grid-template-columns: repeat(auto-fit,
minmax(250px,1fr)); gap:1.8rem; max-width:1100px; margin:0 auto; }
.features .card { background: var(--accent); border-radius: var(--radius);
padding:2rem; box-shadow: var(--shadow); transition: transform 0.3s, box-
shadow 0.3s; }
.features .card:hover { transform: translateY(-8px); box-shadow: 0 15px
25px rgba(0,0,0,0.15); }
.features .card h3 { color: var(--primary); margin-bottom:0.8rem; font-
size:1.2rem; }
.features .card p { color:#333; font-size:0.95rem; }
/* Footer */
footer { background: var(--dark); color:#d4e4f7; text-align:center;

```

```
padding:1.2rem; font-size:0.9rem; }
/* Animations */
@keyframes fadeIn { 0% { opacity:0; transform: translateY(30px); } 100%
{ opacity:1; transform: translateY(0); } }
</style>
</head>
<body>
<header>
  <a href="#" class="logo">🧠 <span>Brain Tumor</span> Portal</a>
  <div class="nav-btns">
    <button
onclick="window.location='patient/patient.html'">Patient</button>
    <button
onclick="window.location='doctor/doctor_login.php'">Doctor</button>
    <button onclick="window.location='admin/login.php'">Admin</button>
  </div>
</header>
<section class="hero">
  <div class="hero-content">
    <h1>Welcome to the <span>Brain Tumor Detection Portal</span></h1>
    <p>Experience AI-powered precision for early tumor detection and
seamless healthcare connectivity.</p>
    <a href="patient/patient.html" class="btn">Get Started</a>
  </div>
</section>
<section id="features" class="features">
  <h2>Our Key Features</h2>
  <div class="grid">
```

```
<div class="card"><h3>🧠 AI Tumor Detection</h3><p>Upload MRI scans and get instant tumor analysis.</p></div>
```

```
<div class="card"><h3>👨‍⚕️ Doctor Consultations</h3><p>Book appointments with verified neurologists easily.</p></div>
```

```
<div class="card"><h3>📄 PDF Health Reports</h3><p>Generate, download, and securely store your medical reports</p></div>
```

```
<div class="card"><h3>⚙️ Admin Dashboard</h3><p>Admin can manage doctors, patients, and appointments </p></div>
```

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